

Research papers

An improved micro–macro model of multicomponent reactive transport for multimineral systems

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ARTICLE INFO

Keywords:

Structure evolution of porous media
Micro-macro model
Multimineral system
Reactive transport
Upscaling

ABSTRACT

The multiscale coupled reactive transport model has attracted increasing attention in recent years because it can reveal the different characteristics of the spatial structure evolution of porous media. A micro–macro model of multiscale coupled reaction transport has been derived from detailed pore-scale models by using the upscaling technique based on homogenization method. Because of its clear upscaling theory, it has been widely used. However, the current micro–macro model still has obvious defects in dealing with the geochemical reaction process in multimineral systems. To solve this problem, we coupled the model with existing geochemical models that have been widely verified so that the model has the ability to deal with multimineral reactions under complex hydrochemical conditions. The newly developed coupling model and the current model are used to simulate the same problem, which proves the superiority of the new model for geochemical reaction prediction. We extend the new model to a two-dimensional scanning electron microscope image, where a challenging medium geometry setup is handled well during simulation. At the same time, the new model also accurately captures the abrupt changes in macroscopic effective parameters caused by the change in geometric topological structure. Finally, we use the principle of mass conservation to quantify numerical errors caused by geometric evolution, ensuring the accuracy and validity of simulation results, which verifies the numerical scheme of the model to a certain extent.

1. Introduction

Reactive transport modeling of porous media has important applications in environmental science as well as energy engineering and engineering geology (Steeff, 2019; Tournassat and Steeff, 2019). For example, reactive transport modeling of porous media can be used to study the transport and conversion processes of organic matter and heavy metals in groundwater, optimize oil and gas production and reservoir injection processes, predict the degradation of concrete structures, etc. Porous media is a medium with a multiscale structure, whose pore-scale ranges from micron to millimeter, and the transport and reaction processes of fluid and matter also involve different time scales (Molins et al., 2019; Steeff et al., 2015; Steeff et al., 2013). Therefore, the numerical simulation and experiments of reactive transport in porous media needs to consider the processes at different scales to accurately describe and predict the migration and chemical reaction

processes of substances in porous media (Ping et al., 2023; Yao et al., 2023; Zhang et al., 2022). In the process of multiscale modeling, upscaling technology is often applied (Weinan, 2011), which can be used to extend the micro model to the macro model. Specific upscaling methods for fluid flow, matter transport, and reaction in porous media include homogenization theory, statistical methods, volume average methods, stochastic methods, etc. (Zhang et al., 2021). With the upscaling technique, it is possible to couple the models at different scales and convert the information from one to another scale to realize the global description and prediction of the mass transfer process in porous media (Scheibe et al., 2015; Tartakovsky et al., 2008; Tomin and Lunati, 2013). For example, at the microscale, the discrete element method (DEM) can be used to describe the process of fluid motion and chemical reaction at the particle scale, while at the macroscale, the continuum hypothesis and finite element method (FEM) can be used to describe the macroscopic transport and chemical reaction of fluid (Wu et al., 2019;

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<https://doi.org/10.1016/j.jhydrol.2024.130764>

Received 10 October 2023; Received in revised form 25 December 2023; Accepted 26 December 2023

Available online 24 January 2024

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Zhang et al., 2022).

In recent years, the micro–macro coupling model developed from homogenization theory has increasingly attracted interest from researchers because it can predict the transport and reaction behavior in porous media at different scales (Gärttner et al., 2020; Ray et al., 2019; Van Noorden, 2009a). A micro–macro model generated by homogenization theory, which can characterize the changing solid–liquid interface. In this model, mineral precipitation and dissolution reactions alter the structure of the porous medium. More precisely, dissolved chemical species are transported by convection and diffusion in porous media and react with minerals that have a nonnegligible volume fraction and density. The micro–macro model consists of flow and transport equations of the concentrations of each species at the continuous scale (macroscale) of the porous medium. Both include averaged time- and space-dependent parameters such as permeability, porosity, reaction surface, and effective diffusion tensor. These parameters are derived from the representative elementary volumes (REVs) with changing geometry and the auxiliary cell problems defined therein (microscale). Changes in the underlying geometry due to dissolution reactions as well as the interface between minerals and liquids are characterized by means of a level-set, which is a classical approach to incorporate evolving interfaces into the model. Van Noorden (2009a); Van Noorden (2009b) presents an extension of the formal two-scale asymptotic expansion, including a level-set description of the evolving solid–liquid interface. The method has been applied to precipitation/dissolution reactions (Lopes et al., 2020; Van Noorden, 2009a; Van Noorden, 2009b; van Noorden et al., 2010), concrete strength degradation (Bourbatache et al., 2012; Bourbatache et al., 2016; Bourbatache et al., 2013), locally periodic media (Bourbatache et al., 2019; Muntean and Nikolopoulos, 2020), biofilm growth (Ebigo et al., 2010; Schulz and Knabner, 2017a; Schulz and Knabner, 2017b), nonisothermal reaction flow (Bringedal et al., 2016a; Bringedal et al., 2016b), slightly compressible subsurface physics (Amanbek et al., 2019), adsorption–desorption processes (Chew and Law, 2020), colloidal kinetics (Ray et al., 2012), drug delivery (Ray et al., 2013), and reactive flow dominated by convection or diffusion (Bourbatache et al., 2020; Kumar et al., 2011; Ray et al., 2019). On the other hand, information from the solution of the macroscopic transport equation is considered to determine the evolution of the solid–liquid interface (micro–macro scale) and the effect of the magnitude and direction of the flow rate on the dissolution pattern. The entire model includes several levels of coupling, such as radial symmetry cases that have been investigated numerically (Bringedal and Kumar, 2017; Schulz et al., 2017; Van Noorden, 2009a), and the main challenge thus far has been the restrictive geometric settings. This results in a greatly simplified set of equations solved by ordinary differential equations instead of the horizontal set equations coupled to the transported partial differential equations. In Ray et al. (2019), the micro–macro problem of complete coupling at a given velocity is considered. Recently, similar models based on the phase-field method, including Darcy flow and two-phase flow, have been numerically studied (Bastidas et al., 2021; Bringedal et al., 2020; Metzger and Knabner, 2021).

Although the micro–macro model has been developed to two-mineral systems, and the treatment of aqueous solution components can also be extended to four components (Gärttner et al., 2022), there are still the following defects: 1) The current method can only deal with simple two-mineral systems, but the real groundwater system is often a complex system with multimineral coexistence; 2) Due to the law of mass action, primary species will exist in groundwater in a certain number of secondary species and participate in chemical reactions in an active form, while the original model simplifies this process and makes primary species directly participate in the reaction. This treatment method will no longer be suitable as the number of minerals increases; 3) The modeling method of micro–macro model in the actual underground multimineral complex system has not been proposed. 4) Although standard level-set methods are widely used and successfully applied to characterize the time-varying fluid domains in reactive

transport simulations (Li et al., 2008; Olsson et al., 2007), but they are not suitable for characterizing the evolution of nonstationary triple points due to direct interactions among multimineral phases. To solve this problem, Gärttner et al. (2022) applied the Voronoi implicit interface method (VIIM), as presented by Frolković et al. (2020) as an extension of the research done by Saye and Sethian (2012). This method constitutes a generalization of the standard level set methods to cover multiphase motion and characterize the physical behavior of nonstationary triple points. The VIIM used in the current model to simulate the evolution of multi-phase interface has only first-order accuracy, and its error magnitude is unknown, which needs to be further quantified to evaluate the accuracy and applicability of the simulation results.

Aiming at the above problems, we proposed a general micro–macro model modeling method, and developed a new micro–macro model of reactive multimineral systems (RTSPHEM-P) (Li, 2023) by coupling the PHREEQC (Parkhurst and Appelo, 2013) based on the RTSPHEM model framework (Gärttner and Frank, 2021). More specifically, in solving the macroscopic equations of the model (RTSPHEM), we construct a micro–macro model of multicomponent reactive transport in a multimineral system by coupling solute transport with geochemical reaction. In Section 2, we introduce the pore-scale model and the micro–macro model after upscaling. In Section 3, we describe numerical methods for solving the microcosmic problems that arise and how to couple the geochemical processes. In Section 4, we use the original model and the newly developed model for numerical simulation of the same problem and demonstrate the capability and advantages of our model by comparing the simulation results. Furthermore, in Section 5, we apply the model to a two-dimensional scanning electron microscopy (SEM) image containing a variety of minerals, prove the practical value of the model through direct numerical simulation, and further evaluate the reliability of the simulation results. In Section 6, we finally discuss the future research direction and application prospects of this model.

2. Mathematical model

Here, we recapitulate a pore-scale model of mineral dissolution reactions and a correspondingly scaled micro–macro model similar to Van Noorden (2009a). We consider the transport and flow of the porous medium on $\Omega \subset \mathbb{R}^2$ in the rectangular domain, with the boundary being $\partial\Omega$. In the context of the micro–macro simulations, an underlying microstructure is attached to each grid point to this macro domain. Assuming an ideal porous medium, the solid microstructure has a positive periodicity of periodicity length $\varepsilon \ll 1$. More precisely, there exists a representative grid of scaled unit cells $Y_\varepsilon := \varepsilon Y := \varepsilon[-\frac{1}{2}, +\frac{1}{2}]^2$, which is assumed to consist of a fluid and a solid phase. Thus, the entire domain Ω is represented as an association of regular ε grids, as shown in Fig. 1 (Gärttner et al., 2022), where:

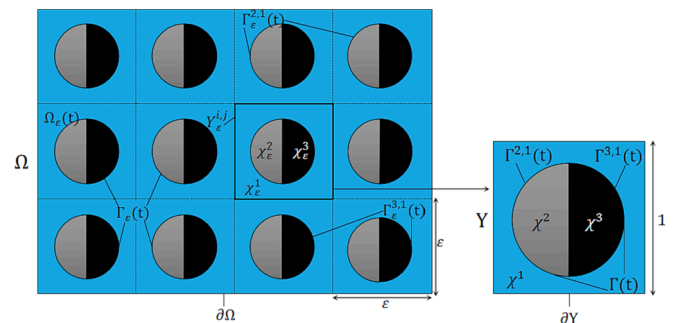


Fig. 1. Geometrical setup: Periodic decomposition with scale parameter ε of domain Ω into the fluid phase χ_ε^1 and the two solid phases $\chi_\varepsilon^2, \chi_\varepsilon^3$ separated by interfaces $\Gamma_\varepsilon^{2,1}$ and $\Gamma_\varepsilon^{3,1}$. Analogous decomposition of related unit cell Y on the right.

$$\Omega = \bigcup_{i,j} Y_{\varepsilon}^{i,j}, \Omega_{\varepsilon} = \bigcup_{i,j} (Y_{\varepsilon}^1)^{i,j} \quad (1)$$

where i and j represent the row and column indexes in the regular grid; ε is a scale separation parameter, represents the ratio of micro and macro length scales of features. Since chemical reactions occur on the solid surface, the liquid–solid interface (internal boundary) captures the time evolution of the system. We further state the following:

$$\Gamma_{\varepsilon} = \bigcup_{i,j} \Gamma_{\varepsilon}^{i,j} \quad (2)$$

2.1. Pore-scale model

In the following, we consider a series of problems that depend on the scaling parameter ε . Therefore, all variables that occur have the subscript ε . To address domains that change over time, the level-set method is applied (Li et al., 2008; Olsson and Kreiss, 2005; Xu et al., 2012). Level sets can characterize the fluid phase (or pore space) of saturated porous media by:

$$L_{\varepsilon}(t, x) = \begin{cases} < 0 & \text{fluid phase / porespace} \\ = 0 & \text{interface} \\ > 0 & \text{solid phase} \end{cases} \quad (3)$$

The level-set L_{ε} variable satisfies a partial differential equation scaled by ε (Osher and Fedkiw, 2003):

$$\frac{\partial L_{\varepsilon}}{\partial t} + v_{n,\varepsilon} |\nabla L_{\varepsilon}| = 0, x \in \Omega \quad (4)$$

where $v_{n,\varepsilon}$ is the normal displacement velocity of the fluid–solid interface. The dissolution/precipitation process that occurs at the fluid–solid interface causes a change in the shape of the solid geometry, so $v_{n,\varepsilon}$ can be related to the surface reaction rate r_{ε} at the interface in a manner of mass conservation:

$$v_{n,\varepsilon} = V_m r_{\varepsilon} \quad (5)$$

where V_m is the molar volume of the mineral.

For a heterogeneous reaction with a reaction rate of r_{ε} , the scale transport equation for the components' concentration c_{ε} is as follows:

$$\frac{\partial c_{\varepsilon}}{\partial t} + \nabla \cdot (c_{\varepsilon} \mathbf{u}_{\varepsilon}) - \nabla \cdot (D \nabla c_{\varepsilon}) = 0, x \in \Omega_{\varepsilon}(t) \quad (6-1)$$

$$(-\mathbf{u}_{\varepsilon} c_{\varepsilon} + D \nabla c_{\varepsilon}) \cdot \mathbf{n}_{\varepsilon} - \varepsilon \rho_s^{-1} r_{\varepsilon} (c_{\varepsilon} - \rho_s) = 0, x \in \Gamma_{\varepsilon}(t) \quad (6-2)$$

$$c_{\varepsilon} = c^0, x \in \Omega_{\varepsilon}(0) \quad (6-3)$$

where D is the molecular diffusion coefficient; $\mathbf{u}_{\varepsilon}$ is the advective velocity field; $\mathbf{n}_{\varepsilon}$ is the unit normal vector of the fluid–solid interface; ρ_s is density of the solid phase. The law of conservation of mass is applied to the internal interface on the assumption that the interface cannot hold the mass constant. As a solution to the ε -scale Stokes problem, the advective velocity field $\mathbf{u}_{\varepsilon}$ and its corresponding pressure field p_{ε} can be obtained:

$$\varepsilon^2 \Delta \mathbf{u}_{\varepsilon} = \frac{1}{\mu} \nabla p_{\varepsilon}, x \in \Omega_{\varepsilon}(t) \quad (7-1)$$

$$\nabla \bullet \mathbf{u}_{\varepsilon} = 0, x \in \Omega_{\varepsilon}(t) \quad (7-2)$$

$$\mathbf{u}_{\varepsilon} = v_{n,\varepsilon} \mathbf{n}_{\varepsilon}, x \in \Gamma_{\varepsilon}(t) \quad (7-3)$$

where μ is the dynamic viscosity. The conditions imposed on $\Gamma_{\varepsilon}(t)$ correspond to the no-slip boundary on the time evolution domain.

2.2. Upscaling

For model upscaling, the volume averaging method described by Whitaker (1999) is widely used. Similarly, the limit $\varepsilon \rightarrow 0$ can be formally identified by using the two-scale asymptotic expansion method. The relevant mathematical description can be found in the studies of Allaire (1992), Auriault et al. (2009), and Bensoussan et al. (1978). Concerning the scale separation, in addition to the global variable x , the micro variable y is also introduced. The two are related by the relation $y = x/\varepsilon$. Consequently, the spatial gradient, divergence, and Laplacian operators expand as follows:

$$\nabla = \nabla_x + \frac{1}{\varepsilon} \nabla_y, \nabla \bullet = \nabla_x \bullet + \frac{1}{\varepsilon} \nabla_y \bullet, \Delta = \Delta_x + \frac{2}{\varepsilon} \nabla_x \bullet \nabla_y + \frac{1}{\varepsilon^2} \Delta_y \quad (8)$$

Furthermore, we assume that all variable functions can be expanded into an asymptotic series of scaling parameters ε , i.e.:

$$\varphi_{\varepsilon}(t, x) = \varphi_0(t, x, y) + \varepsilon \varphi_1(t, x, y) + \varepsilon^2 \varphi_2(t, x, y) + \dots \quad (9)$$

In addition to the standard expansion introduced above in the level-set description framework, the level-set L_{ε} itself and the unit normal vector must also be expanded due to the evolution of the microstructure. For a two-dimensional setup, a normal expansion can be expressed as a level-set, resulting in the following expression (Van Noorden, 2009a):

$$L_{\varepsilon}(t, x) = L_0(t, x, y) + \varepsilon L_1(t, x, y) + \varepsilon^2 L_2(t, x, y) + \dots \quad (10-1)$$

$$\mathbf{n}_{\varepsilon} = \mathbf{n}_0 + \varepsilon \mathbf{n}_1 + O(\varepsilon^2) \quad (10-2)$$

$$\mathbf{n}_0 = \frac{\nabla_y L_0}{|\nabla_y L_0|}, \mathbf{n}_1 = \tau_0 \frac{\nabla_x L_0 + \nabla_y L_1}{|\nabla_y L_0|} \quad (10-3)$$

where τ_0 is the unit tangent vector ($\tau_0 \perp \mathbf{n}_0$) and $O(\varepsilon^2)$ represents the higher order infinitesimal of ε^2 . When applying these expansions to our problem and analyzing the different orders of ε , one can obtain the desired description of the macroscopic homogenization problem, i.e., the micro–macro model. Note that the mathematical principle of the single mineral case is similar to that of the multimaterial case, and the equations of the multimaterial case are too complicated in terms of mathematical model description. For ease of understanding, the description of the mathematical model is given here in the form of a single mineral as an example. The mathematical model description of multimaterial can be referred to the work of Gärtner et al. (2022).

2.3. Micro-macro model

The micro–macro model of mineral reactions results as an asymptotically expanded leading-order approximation of the pore-scale model (Van Noorden, 2009a), i.e., in order 0, which is indicated in the following using the subscript 0. The time-dependent liquid phase domain at the microscopic scale is represented as $Y_{l,0}(t, x) := \{y \in Y: L_0(t, x, y) < 0\}$, and the solid–liquid interface is $\Gamma_0(t, x) := \{y \in Y: L_0(t, x, y) = 0\}$. L_0 is propagated by the level-set equation:

$$\frac{\partial L_0}{\partial t} + v_{n,0} |\nabla L_0| = 0, \text{in } \Omega \times Y \quad (11)$$

The reaction transport equation of the component concentration c_0 is as follows:

$$\frac{\partial \phi c_0}{\partial t} + \nabla_x \cdot (\bar{\mathbf{u}}_0 c_0) - \nabla_x \cdot (\mathbf{D} \nabla c_0) = -s r_0, \text{in } \Omega \quad (12)$$

where $\bar{\mathbf{u}}_0$ is the average velocity field; $\mathbf{D}$ is the effective diffusion tensor; ϕ is the porosity; and s is the reactive specific surface area. The calculation formulas for ϕ and s are as follows:

$$\phi = \frac{|Y_{l,0}|}{|Y|}, s = \frac{|\Gamma_0|}{|Y|} \quad (13)$$

The average velocity field $\bar{u}_0$ is calculated by Darcy's law and the continuity of the fluid:

$$\bar{u}_0 = -\frac{\mathbf{K}}{\mu} \nabla p_0, \text{ in } \Omega \quad (14-1)$$

$$\nabla \bullet \bar{u}_0 = 0, \text{ in } \Omega \quad (14-2)$$

where $\mathbf{K}$ is the permeability tensor.

The diffusion tensor and permeability tensor are defined by the solution of the standard auxiliary cell problems (or REV) (Hornung, 1997; Van Noorden, 2009a), and the diffusion tensor $\mathbf{D}$ is defined as follows:

$$\mathbf{D}_{ij}(t, x) := \frac{1}{|Y|} \int_{Y_{l,0}(t, x)} D(\partial_{y_i} \zeta_j + \delta_{ij}) dy, i, j \in 1, 2 \quad (15)$$

where δ_{ij} is the Kronecker delta and ζ_j is the solution to the auxiliary cell problems, which represents the concentration disturbance field. The solution is as follows:

$$-\nabla_y \bullet (\nabla_y \zeta_j) = 0, \text{ in } Y_{l,0}(t, x) \quad (16-1)$$

$$\nabla_y \zeta_j \bullet \mathbf{n}_0 = -e_j \bullet \mathbf{n}_0 \text{ on } \Gamma_0(t, x) \quad (16-2)$$

$$\zeta_j \text{ periodic in } y, \frac{1}{|Y|} \int_{Y_{l,0}} \zeta_j dy = 0 \quad (16-3)$$

where e_j is the unit vector. The permeability tensor $\mathbf{K}$ is defined as follows:

$$\mathbf{K}_{ij}(t, x) := \frac{1}{|Y|} \int_{Y_{l,0}(t, x)} \omega_j^i dy, i, j \in 1, 2 \quad (17)$$

where ω_j is the solution of the auxiliary cell problems, which can be considered the source of the velocity space deviation, and we call it the velocity disturbance field. Here, we introduce another vector π_j , which can be considered the source of the pressure space deviation, which we call the pressure disturbance field. The solution is to solve the auxiliary cell problems of Stokes type (Hornung, 1997):

$$-\Delta_y \omega_j + \nabla_y \pi_j = e_j, \text{ in } Y_{l,0}(t, x) \quad (18-1)$$

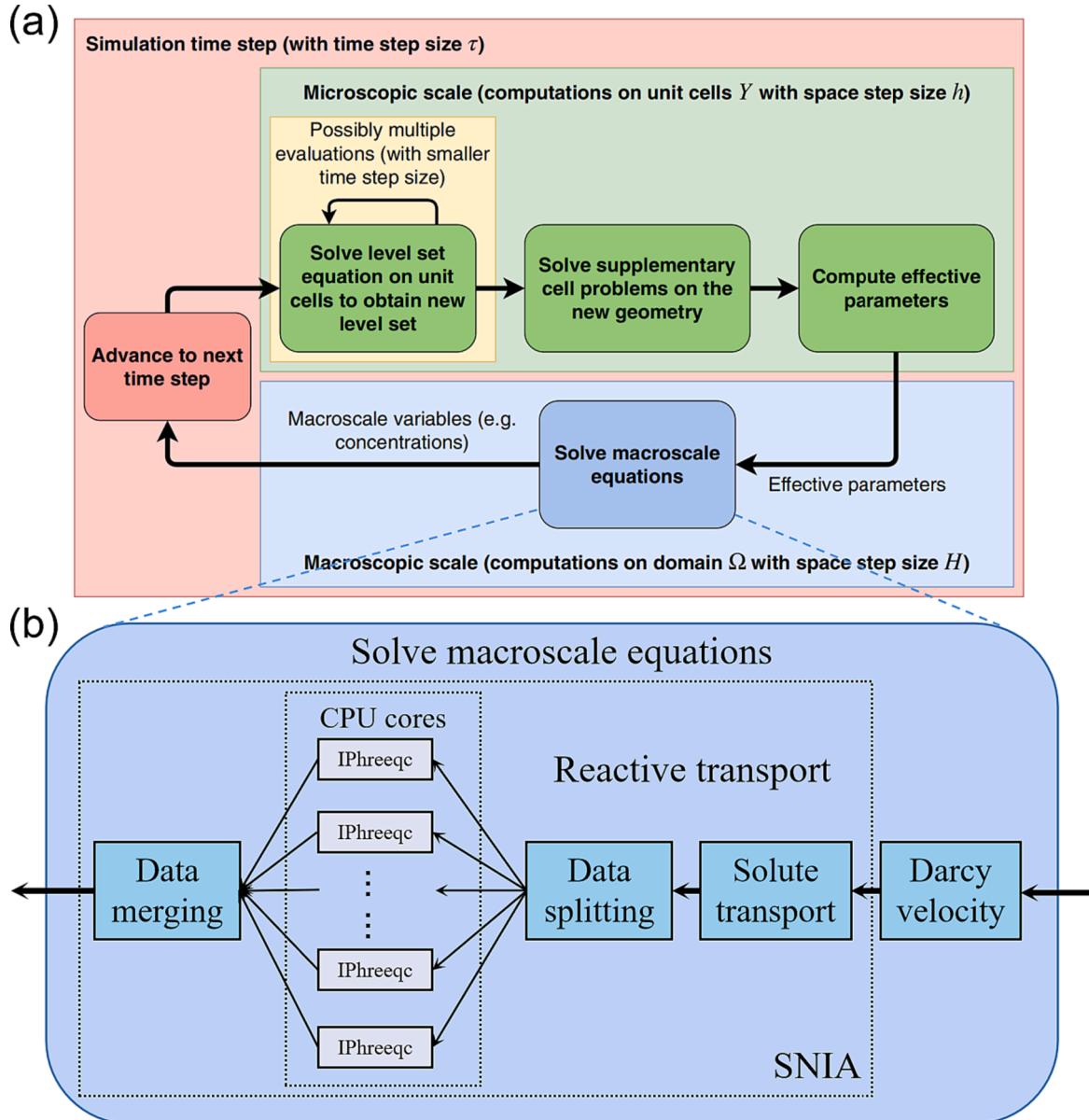


Fig. 2. The frame structure of the RTPHEM-P model. (a) The overall framework of the model; (b) The macromodel calculation module.

$$\nabla_y \bullet \omega_j = 0 \text{ in } Y_{l,0}(t, x) \quad (18-2)$$

$$\omega_j = 0, \text{ on } \Gamma_0(t, x) \quad (18-3)$$

$$\omega_j, \pi_j \text{ periodic in } y, \frac{1}{|Y|} \int_{Y_{l,0}} \pi_j dy = 0 \quad (18-4)$$

3. Numerical implementation

The data exchange between the micro and macro scales uses a simple operator separation method, which can connect the transport/flow solver with the cell problem solver through the two-way transmission of data. This approach is commonly used in multiscale simulations (Bastidas et al., 2021; Ray et al., 2019). The complete algorithm of the model is illustrated in Fig. 2a (Ray et al., 2019). All numerical implementations are implemented under the framework of MATLAB R2021b (MATLAB, 2021) and the PDE toolbox RTSPHEM (Gärttner and Frank, 2021). The original RTSPHEM model framework simplified the geochemical reaction, using the total concentration of the primary species to directly participate in the reaction, and adopted the global implicit approach (GIA) (Steefel et al., 2014; Yeh and Tripathi, 1989) to solve the reaction transport process. This approach is not in line with the actual situation. To solve a more complete geochemical reaction, IPHreeqc is coupled into RTSPHEM by the sequential noniterative approach (SNIA) (Steefel and MacQuarrie, 1996) (Fig. 2b). More precisely, we further optimize and extend the RTSPHEM model framework and develop the RTSPHEM-P model by using the programming technique of CPU parallelization, which can be applied to solve the micro-macro problems of multi-mineral reaction systems proposed in this paper. In addition, it should be noted that when developing the RTSPHEM-P model, we shielded the segment of the RTSPHEM code that was originally responsible for calculating geochemical reactions. The discretization details of space and time in the RTSPHEM-P model can be found in the studies of Gärttner et al. (2020); Gärttner et al. (2022) and Ray et al. (2019).

4. Comparative verification of models

Below, we apply a microcosmic model to the dissolution of heterogeneous mineral structures. Again, we provide all information in dimensional form. To provide a realistic and challenging example, we base the initial basic geometry on data provided by the 10th SPE Comparative Solution Program (SPE10) (Christie and Blunt, 2001). These data are composed of detailed permeability and porosity, from which we derive the initial basic geometric background. We use the RTSPHEM model and the RTSPHEM-P model to simulate the same scenario and verify the reliability and superiority of the new model by comparing the results.

4.1. Model settings

We consider the macro domain $\Omega = 22 \times 6 \text{ mm}^2$ for the micro-macro simulation below. In the initial phase, we want the permeability distribution to be similar to that given at the top level of the permeability map of the SPE10 data (Fig. 3a). We cannot use these permeability data directly because all valid parameters come from the representative geometry in our framework. In the first step, the macro information is thus transformed into the simulated initial configuration of the representative cell geometry. To reduce the overall complexity of our simulation, each coarse-up data block covers an average of 5×5 original data blocks, which constitutes 44×12 domain decompositions (Fig. 3c). Each subdomain will be associated with a cell, i.e., each representative elementary volume (REV) representing the properties of a 0.25 mm^2 macroscopic region. The macro domain itself is discretized using 11,162 triangular elements (Fig. 3b), and each REV controls the macroscopic effective parameters of approximately 20 triangular elements on average. Assuming the underlying microstructure is rectangular, using two degrees of freedom, the length and width of the rectangle (Fig. 3d) are used to adjust the desired initial permeability distribution. Thus, the geometry we derive is upgraded approximately to SPE10 data at the initial time but evolves according to the reactivity migration and flow problems described in the model.

In addition, assume that half of the solid is composed of calcite and

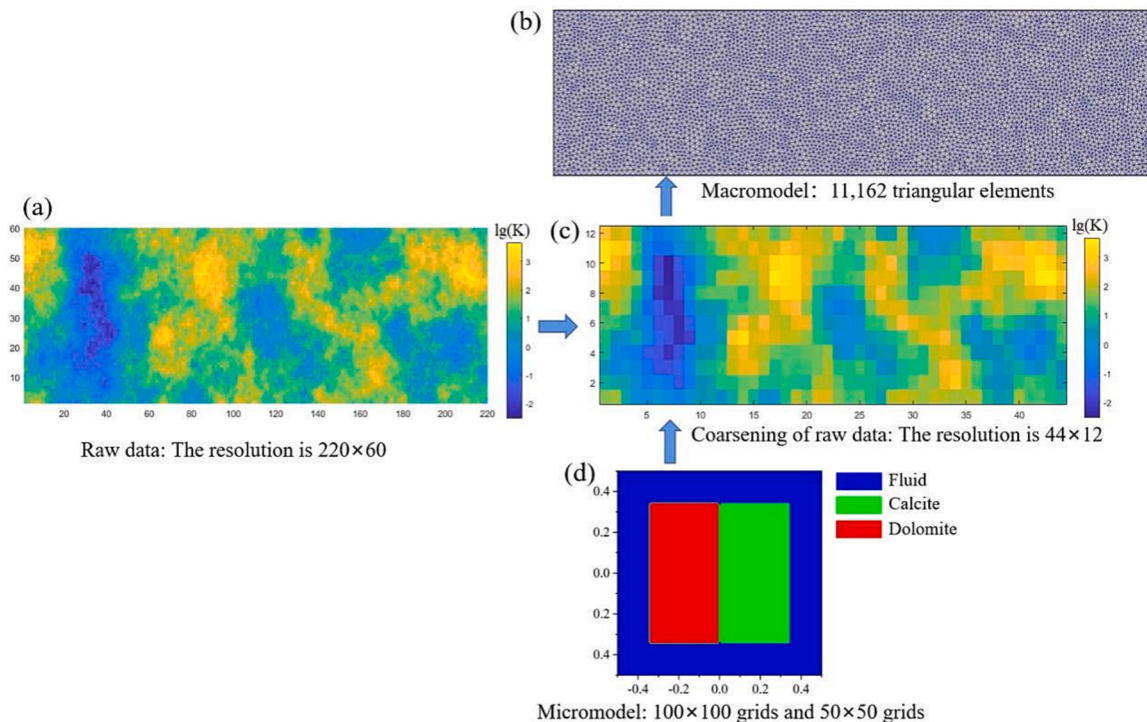
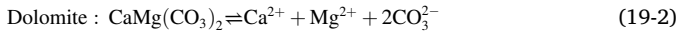


Fig. 3. Grid generation of the model.

the other half is composed of dolomite (Fig. 3d). Accordingly, the reaction path is built into the RTSPHEM model, as follow:



It is not difficult to find that the above chemical reaction equation does not include the consumption of H^+ in the mineral dissolution process. At the same time, we find that the consumption of H^+ in Gärtner et al. (2020) and Gärtner et al. (2022) is extra, and only the consumption of H^+ in the dissolution of calcite is considered, which is not practical. Therefore, we believe that a more complete chemical reaction process should also include the following water-phase equilibrium reactions (Ball and Nordstrom, 1991):



These additional chemical reactions are considered in the RTSPHEM-P model, which can also be handled if further ion exchange and surface coordination reactions need to be considered, and the activity of the reaction components is naturally considered in the RTSPHEM-P model.

The relevant mineral dissolution reaction rate is:

$$r_{\text{Calcite}} = (k_1 c_{\text{H}^+} + k_2) \bullet (1 - \frac{c_{\text{Ca}^{2+}} c_{\text{CO}_3^{2-}}}{K_{\text{Calcite}}}) \quad (21-1)$$

$$r_{\text{Dolomite}} = k c_{\text{H}^+}^{0.5} \bullet (1 - \frac{c_{\text{Ca}^{2+}} c_{\text{Mg}^{2+}} c_{\text{CO}_3^{2-}}^2}{K_{\text{Dolomite}}}) \quad (21-2)$$

where $k_1 = 0.89 \text{ mol/m}^2/\text{s}$, $k_2 = 6.7 \times 10^{-7} \text{ mol/m}^2/\text{s}$, $k = 4.5 \times 10^{-4} \text{ mol/m}^2/\text{s}$, $K_{\text{Calcite}} = 10^{-8.234}$, and $K_{\text{Dolomite}} = 10^{-16.5}$ (Molins et al., 2017; Ray et al., 2019). Here, we define the reaction rate equations of two minerals by modifying the PHREEQC database.

Through the above process, the macroregion is equipped with 528 cells, each cell is geometrically evolved through a 100×100 grid, and the effective parameters K and D of the tensor are calculated through a 50×50 grid, as described in Eqs (15) to (18). The different parts of the outer boundary are defined as Γ_{left} , Γ_{right} , Γ_{up} , and Γ_{down} ; then, the initial and boundary conditions of the RTSPHEM model and the RTSPHEM-P model are:

$$(c^* \vec{u} - D \nabla c^*) \bullet \mathbf{n} = 0 \text{ on } \Gamma_{\text{up}} \text{ and } \Gamma_{\text{down}} \quad (22-1)$$

$$\text{RTSPHEM}^* \in \{\text{Ca}^{2+}, \text{Mg}^{2+}, \text{CO}_3^{2-}, \text{H}^+\}, \text{RTSPHEM.P}^* \in \{\text{Ca}, \text{Mg}, \text{C}, \text{H}^+\} \quad (22-2)$$

$$(c_{\#} \vec{u} - D \nabla c_{\#}) \bullet \mathbf{n} = 0 \text{ on } \Gamma_{\text{left}} \quad (22-3)$$

$$\text{RTSPHEM}(\#) \in \{\text{Ca}^{2+}, \text{Mg}^{2+}, \text{CO}_3^{2-}\}, \text{RTSPHEM.P}(\#) \in \{\text{Ca}, \text{Mg}, \text{C}\} \quad (22-4)$$

$$(c_{\text{H}^+} \vec{u} - D \nabla c_{\text{H}^+}) \bullet \mathbf{n} = c_{\text{H}^+} \vec{u} \bullet \mathbf{n} \text{ on } \Gamma_{\text{left}} \quad (22-5)$$

$$D \nabla c_{\text{H}^+} \bullet \mathbf{n} = 0 \text{ on } \Gamma_{\text{right}} \quad (22-6)$$

$$c_{\#} = 0 \text{ in } \Omega \quad (22-7)$$

$$c_{\text{H}^+} = 10^{-2} [\text{mol/m}^3] \text{ in } \Omega \quad (22-8)$$

where $\{\text{Ca}^{2+}, \text{Mg}^{2+}, \text{CO}_3^{2-}, \text{H}^+\}$ represents the secondary component; $\{\text{Ca}, \text{Mg}, \text{C}\}$ represents the primary component; and $\mathbf{n}$ is the boundary unit normal vector. In addition, the boundary conditions for Darcy's equation are set as follows:

$$\vec{u} \bullet \mathbf{n} = -5 \times 10^{-5} [\text{m/s}] \text{ on } \Gamma_{\text{left}} \quad (23-1)$$

$$\vec{u} \bullet \mathbf{n} = 0 \text{ on } \Gamma_{\text{up}} \text{ and } \Gamma_{\text{down}} \quad (23-2)$$

$$p = 0 \text{ on } \Gamma_{\text{right}} \quad (23-3)$$

Since the RTSPHEM-P model uses the SNIA scheme to solve the reaction transport process, the size of the time step needs to consider the Courant number ($\text{Cr} = \Delta t \bullet \vec{u} / \Delta x \leq 1$), so we set the maximum time step of the model to 4 s and the total simulation time to 4.32×10^3 s. Since the GIA scheme is used in the RTSPHEM model, the maximum time step is set to 10,000 s and the total simulation time is set to 4.32×10^7 s without considering the size of the Courant number. If the RTSPHEM-P model uses the same simulation duration as the RTSPHEM model, the corresponding number of time steps will be 10,000 times that of the RTSPHEM model, which is an unacceptable computational cost. Therefore, we first simulate the time required for the reaction to reach the stable stage (≥ 1000 s) using the RTSPHEM-P model, and the longer simulation time (4.32×10^6 s) is done by scaling.

4.2. Comparison of the simulation results

Since the RTSPHEM model does not consider other aqueous equilibrium reactions after the dissolution of calcite and dolomite, the concentration of calcium ions, carbonate ions, and magnesium ions in the fluid can naturally represent the total concentration of calcium, carbon, and magnesium in the fluid, and the hydrogen ions represent the total concentration of hydrogen in addition to water molecules. In the RTSPHEM-P model, we use the total concentration of calcium, carbon, and magnesium as the principal variable of transfer between reaction and transport, and the activity of hydrogen ions is used. In this way, the physical quantities that are represented by calcium ions, carbonate ions, magnesium ions, and hydrogen ions in the RTSPHEM model can be correspondingly consistent with the RTSPHEM-P model, and thus, the two are comparable models.

Fig. 4 shows the difference between the two models of the calcium concentration in the simulation results. The results show that the reaction reaches the stable stage, the distribution range of the calcium concentration in the RTSPHEM model is $1.9 \times 10^{-6} \sim 6.9 \times 10^{-5} \text{ mol/L}$. Accordingly, the distribution of the calcium concentration in the RTSPHEM-P model ranges from 3.2×10^{-6} to $6.4 \times 10^{-5} \text{ mol/L}$. Combined with the coloring of Fig. 4a and 4b, it is not difficult to see that the simulation result of the calcium concentration by the RTSPHEM model is higher than the simulation result of the calcium concentration by the RTSPHEM-P model. This is mainly because the consumption of hydrogen ions in the dissolution process of dolomite is not considered in the RTSPHEM model, which leads to a high concentration of hydrogen ions, and the concentration of hydrogen ions is positively correlated with the dissolution rate of calcite and dolomite (Eq. (19)), so the concentration of calcium is correspondingly high. In addition, since the aqueous equilibrium reaction also consumes hydrogen ions, the dissolution rate of calcite and dolomite in the RTSPHEM-P model is further reduced. Furthermore, the carbon concentration simulation results are similar to the calcium concentration simulation results, which is related to the reaction path of calcite and dolomite, both of which dissolve to

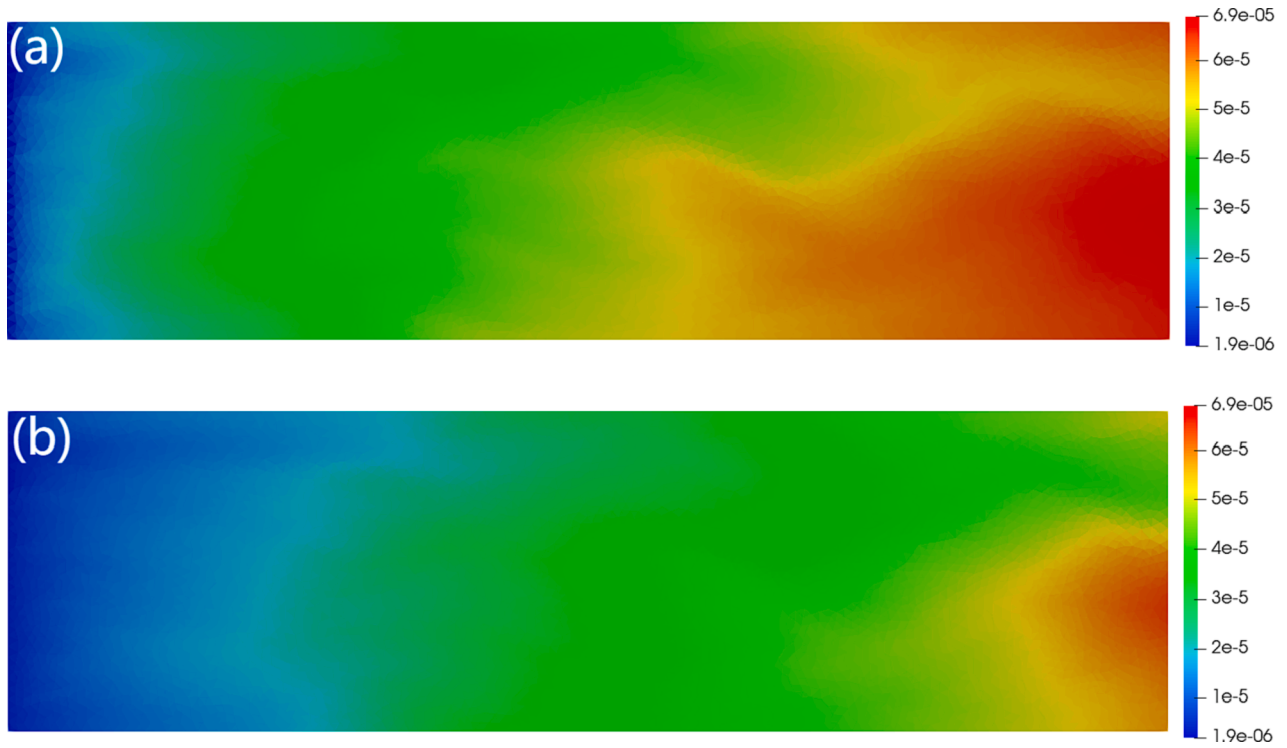


Fig. 4. Simulation results after 4320 s. (a) Calcium ion concentration distribution in the RTSPHEM model [mol/L], and (b) total calcium concentration distribution in the RTSPHEM-P model [mol/L].

produce calcium and carbon, and a demonstration of the simulation results of both models of the carbon concentration can be found in the [supporting materials](#).

Since the concentration of hydrogen ions in the RTSPHEM model is fully involved in the reaction, its concentration can be considered as activity; in other words, its concentration can be expressed by the pH

value ($-\lg[H^+]$) ($c_{H^+} = 10^{-2} \text{ mol/m}^3$ is equal to $\text{pH} = 5$). As shown in [Fig. 5](#), the pH distribution range in the RTSPHEM model is $5.1 \sim 16$, and correspondingly, the pH distribution range in the RTSPHEM-P model is $5.5 \sim 9.5$. The normal pH range is 0 to 14, and the upper limit of the lower range of the RTSPHEM model exceeds 14, which is not in line with the actual situation. This further proves that the simplification of the

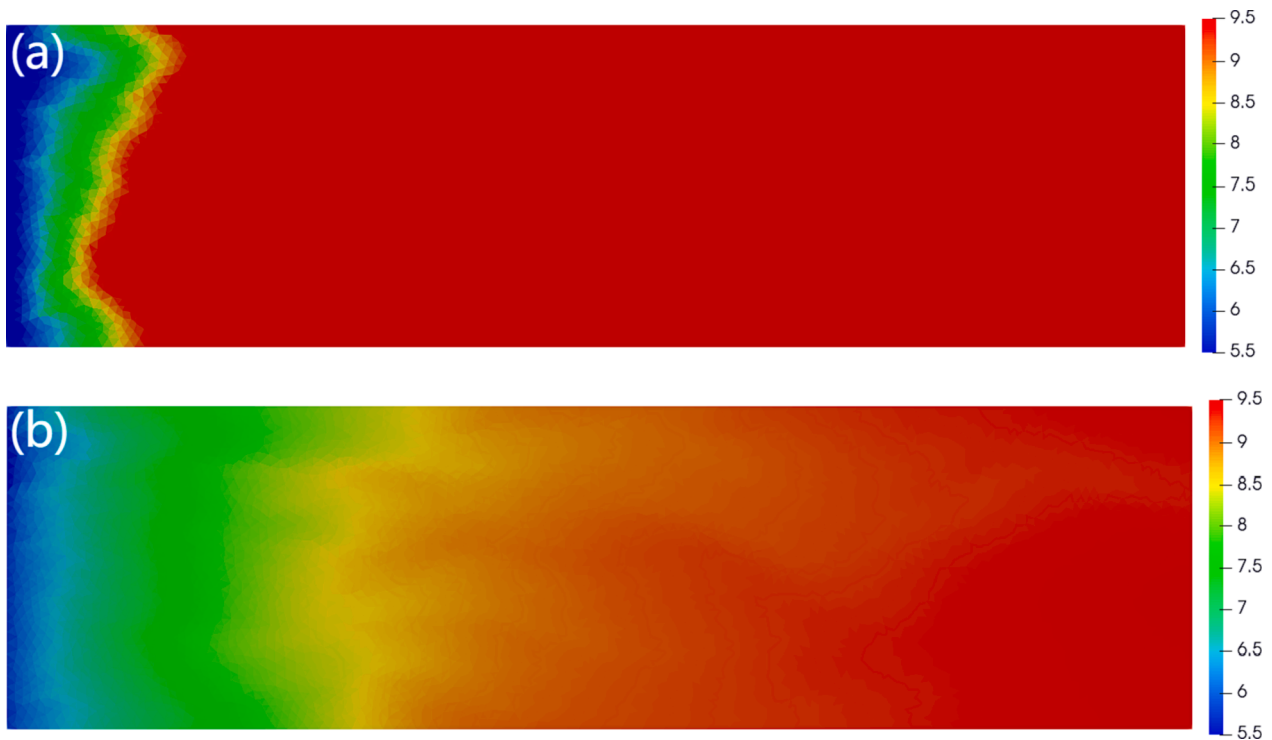


Fig. 5. Simulation results after 4320 s. (a) The pH distribution of the RTSPHEM model and (b) the pH distribution of the RTSPHEM-P model.

chemical reaction model is unreasonable and will lead to a distortion of the simulation results. In contrast, the simulation results of the pH value in the RTSPHEM-P model are more reasonable, which agrees with our consistent understanding of hydrogeochemistry. In addition, it can be seen from the reaction path of dolomite that the concentration of hydrogen ions is the main factor controlling its reaction rate, so the distribution of the magnesium concentration is also bound to be closely related to the distribution of the pH value. From the simulation results, the distributions of the magnesium concentration and the pH are similar (see the [supplementary material](#)).

To further investigate the formation of preferential flow channels during the dissolution process, we investigate the evolution model of the Darcy flow field during the long period of reaction migration. Due to the limitation of the Courant number, it is difficult for the RTSPHEM-P model to maintain the same simulation duration as the RTSPHEM model under the current inlet flow rate (5×10^{-5} m/s), so it is necessary to scale the simulation duration of the RTSPHEM-P model. Here, we have amplified the reaction rate of the RTSPHEM-P model by a factor of 1000 to enlarge the equivalent simulation time by a factor of 1000. Therefore, significant changes in the flow field are observed over a period of 4.32×10^3 s (equivalent to 4.32×10^6 s) for the total duration of the simulation. As shown in Fig. 6, the simulation results of the scaled RTSPHEM-P model are the same as those of the RTSPHEM model, capturing the formation of priority flow channels. In addition, the locations of the two models' priority channels are the same, and they are close to the upper boundary of the model. In fact, this linear scaling

necessarily violates the conservation of mass if the fluid velocity (or some other parameters of the model) is not properly adjusted. The errors caused by the scaling process mainly come from two aspects, one is the insufficient precision of the fluid–solid interface characterization method, and the other is the nonlinear kinetic process of the minerals' dissolution reaction. Our aim in doing so is to be able to demonstrate the simulation capabilities of the improved model in capturing the formation process of priority flow channels with a shorter time and smaller computational effort. The main way to reduce this error without changing the initial inflow velocity is to change the average size of the macro grid, or to extend the simulation time. Here, our main purpose is to qualitatively evaluate the function of the model, so only the consistency of the grid is held in the comparison process, and mass conservation is not considered. Although there is some subjectivity in the model setting, the simulation results show that the model has a good performance in describing the formation process of priority flow channels in multimineral reaction systems.

In summary, the simulation results of the RTSPHEM-P model are closer to the actual situation of reactive transport during the dissolution of carbonate minerals than the simulation results of the RTSPHEM model and have good practical value. The successful development of the RTSPHEM-P model makes the micro–macro coupling model, which is highly theoretical, capable of simulating real scenes at the field-scale or core-scale. Therefore, in the next chapter, we apply the RTSPHEM-P model to practical applications in real scenarios. In addition, both models are run on a computer with an Intel Core i9-12900 K CPU, which

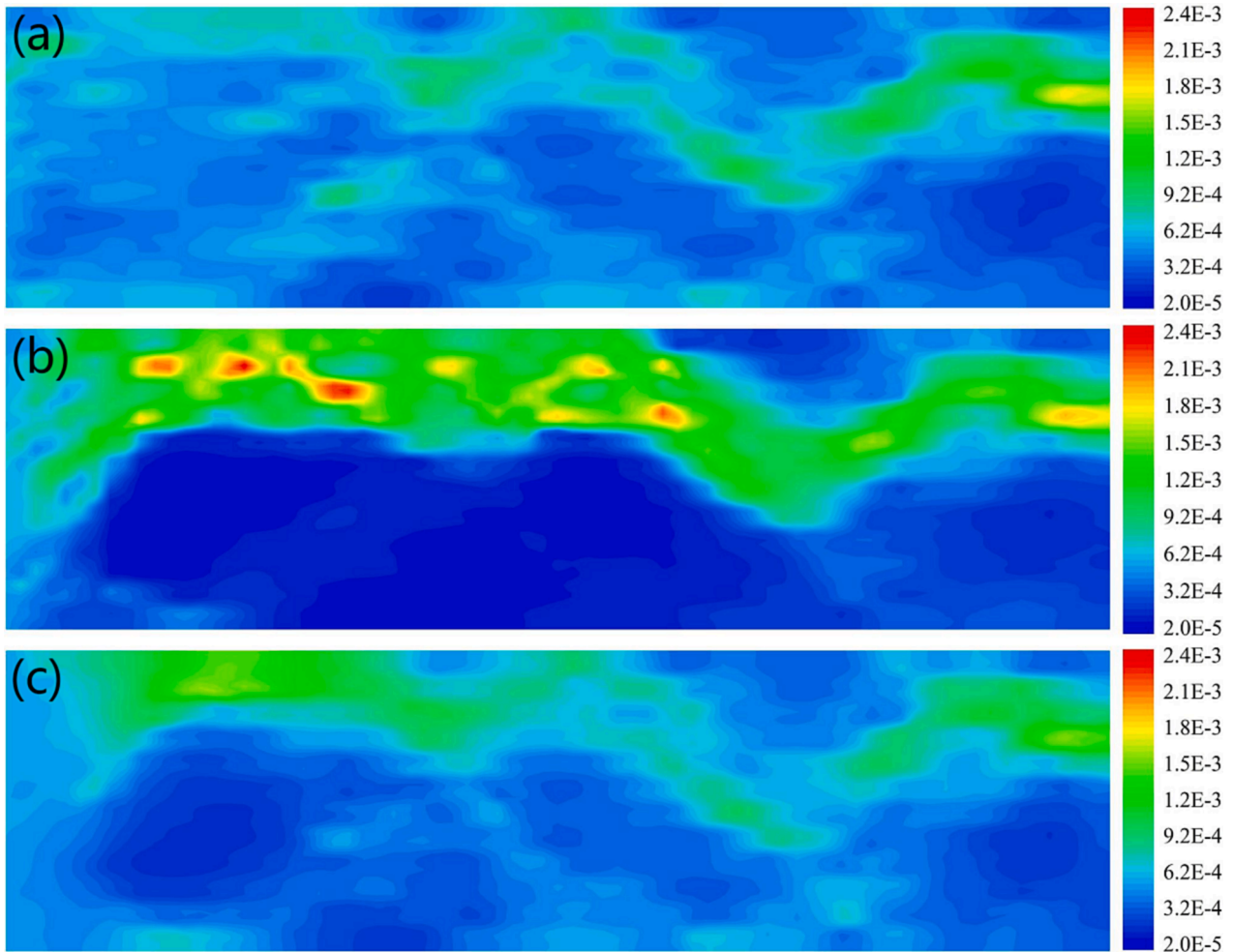


Fig. 6. (a) Initial flow field velocity (dm/s); (b) RTSPHEM model flow field velocity and simulation results; (c) RTSPHEM-P model flow field velocity and simulation results after 4.32×10^3 s. Note that the value refers to the magnitude of the velocity, not the x-component.

has 16 physical cores, and the overall computational time achieved by the RTSPHEM model is approximately 63,000 s, and the overall computational time achieved by the RTSPHEM-P model is approximately 21,000 s. Both models occupy less than 32 GB of memory during runtime.

5. Generalized application of the model

Below, we apply the improved microscope-macro (RTSPHEM-P) model to more complex problems. Fortunately, the study of Ma et al. (2021) provides a realistic and challenging example. Ma et al. used a large 2-dimensional scanning electron microscopy (SEM) image (resolution 9474×6947 , pixel size $1.2 \mu\text{m}$) to investigate the relationship between the mineral's accessible surface area and the fluid-mineral reactivity at pore-scale. If the grids were divided according to their original simulation scheme (directly by resolution), the number of grids would be 65, 815, 878. The computational cost of such a grid number is unacceptable, so they only intercepted part of the image (984×870) for performing relevant simulation studies. Given the upscaling characteristics of the micro-macro model, we believe it is very suitable for simulating this large-scale problem. Not only can the micro-macro model solve the macro problem, but the model also considers the micro details, and the computational cost is also easily acceptable.

5.1. Model settings

First, the SEM images are binarized according to Ma et al. (2021) (Fig. 7a), and the permeability of all solid (mineral phase) regions is set at 10^{-16} m^2 , while the permeability of the pore regions is set at 10^{-12} m^2 (Fig. 7c). The construction of the typical units considers the state of bonding between the solid minerals (Fig. 7b), and according to the volume fraction of each mineral in Fig. 7a, we approximate the solid volume fraction of quartz, dolomite, K-feldspar, muscovite, and kaolinite to 37.5 %, 25 %, 15 %, 11.25 %, and 11.25 %, respectively (this refers to the volume fraction that occupies the solid skeleton). The

mineral geometry setting in Fig. 7b is not unique. To fully represent the complex mineral distribution in Fig. 7a, it is necessary to consider the volume fraction of minerals, the fractal geometry, and the accessibility of the reaction fluids at the same time, which will be systematic and complicated work. Therefore, the geometric setting in Fig. 7b is largely designed to implement the functionality of the model. This geometric setting of minerals is a special case, which can reflect that the model can handle not only simple mineral composition but also complex mineral structure composition, so it has a certain representativeness. In addition, this geometric setting can lead to abrupt changes in permeability in the later stage of simulation, which can better show the advantages of the model in capturing abrupt changes of macro parameters.

For the micro-macro simulation detailed below, we consider the macro domain $\Omega = 11.28 \times 8.16 \text{ mm}^2$ ($9,300 \times 6,900$ resolution). After coating, each data block in Fig. 7d covers an average of 300×300 original data blocks, which constitutes a 31×23 domain decomposition (Fig. 7d). Each subdomain will be associated with a cell, i.e., each cell representing the properties of a macroscopic region of 0.0576 mm^2 is consistent with the order of a typical cell body in natural rocks. The macroregion is equipped with a total of 713 typical cell grids, as shown in Fig. 7b. Each typical cell has the same shape, but the internal solid volume is different. The size of the solid volume is related to the permeability, which also determines the porosity of each typical cell. Each typical cell is geometrically evolved with a 100×100 grid, and the effective parameters $\mathbf{K}$ and $\mathbf{D}$ of the tensor are calculated with a 50×50 grid. The use of two grid strategies is to reduce the computational cost of the microscopic model, of course, this handling method is carried out under the premise of ensuring accuracy. The macro domain itself is discretized using 13,778 triangular elements (Fig. 7e), with each typical cell controlling macroscopic effective parameters of on average approximately 20 triangular elements.

Five solid minerals are considered in the model, namely, quartz, dolomite, K-feldspar, muscovite, and kaolinite (because the ilmenite content is very low, it is ignored). Accordingly, the reaction pathways of these minerals are as follows:

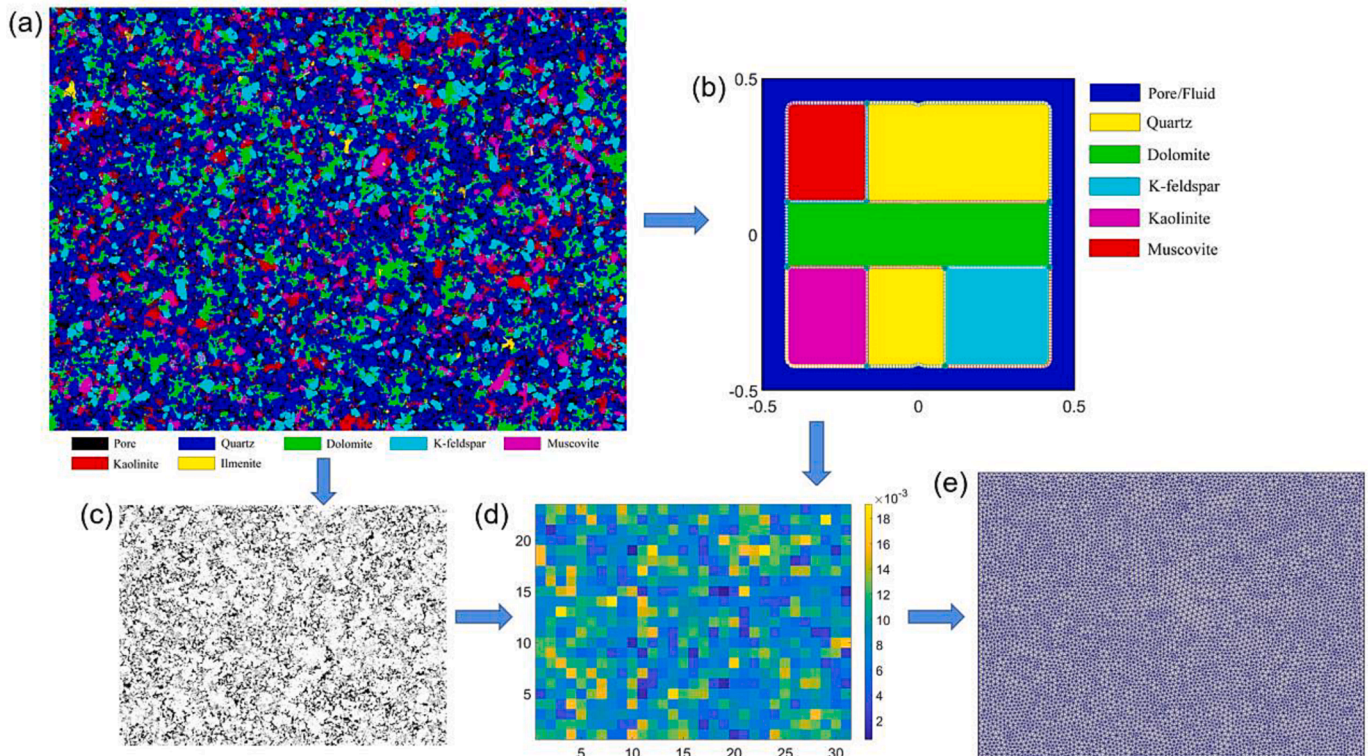
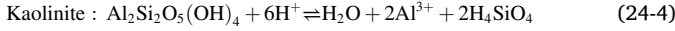
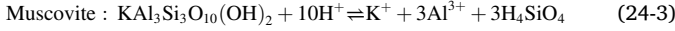
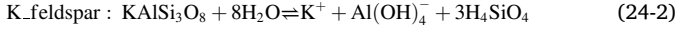


Fig. 7. Preprocessing and mesh generation of the model.



For the dissolution precipitation reaction of minerals, we adopt a more general form. In general, mineral dissolution precipitation is described as a reaction kinetic process, and the rate r is explicitly described as a function of species concentration in the following form (Lasaga, 1984; Steefel and Lasaga, 1994):

$$r_m = \pm k_m \left| 1 - \left(\frac{Q_m}{K_m} \right)^p \right|^q \quad (25)$$

where m is the kinetic index of the mineral reaction. When r_m is positive, dissolution occurs, and when r_m is negative, precipitation occurs. The constant k_m is the inherent reaction rate ($\text{mol}/\text{m}^2/\text{s}$), which is temperature dependent. The constant K_m is the reaction equilibrium; Q_m is the activity product. The parameters p and q are generally determined by experiment and typically, but not always, they are equal to 1. For many minerals, the kinetic rate constant k_m includes all three mechanisms (Lasaga et al., 1994; Palandri and Kharaka, 2004):

$$k_m = k_{25}^{\text{nu}} \exp \left[\frac{-E_a^{\text{nu}}}{R} \left(\frac{1}{T} - \frac{1}{298.15} \right) \right] + k_{25}^{\text{H}} \exp \left[\frac{-E_a^{\text{H}}}{R} \left(\frac{1}{T} - \frac{1}{298.15} \right) \right] a_{\text{H}}^{n_{\text{H}}} + k_{25}^{\text{OH}} \exp \left[\frac{-E_a^{\text{OH}}}{R} \left(\frac{1}{T} - \frac{1}{298.15} \right) \right] a_{\text{OH}}^{n_{\text{OH}}} \quad (26)$$

where E_a is the activation energy (KJ/mol); k_{25} is the reaction rate constant at 25 °C; and the superscripts nu, H, and OH represent the neutral, acid, and base mechanisms, respectively; a is the activity of species (mol/m^3); n is a power term (constant); R is the gas constant; and T is the absolute temperature (K). Since this simulation assumes an isothermal environment (i.e., a temperature of 25 °C or 298.15 K), Eq. (26) can be simplified as:

$$k_m = k_{25}^{\text{nu}} + k_{25}^{\text{H}} \bullet a_{\text{H}}^{n_{\text{H}}} + k_{25}^{\text{OH}} \bullet a_{\text{OH}}^{n_{\text{OH}}} \quad (27)$$

The values of each parameter in the Eq. (27) are shown in Table 1 (Palandri and Kharaka, 2004; Xu et al., 2004).

The solution precipitation reaction paths mentioned above, as well as the more complex water phase equilibrium reactions, have been pre-defined in the PHREEQC database, and the kinetic reaction rate equation has been rewritten in the database. In addition, considering $\text{Cr} \leq 1$, we set the maximum time step of the model to 1000 s and the total simulation time to 1.296×10^6 s.

The initial and boundary conditions of the model are as follows:

$$(c_{\#} \vec{u} - \mathbf{D} \nabla c_{\#}) \bullet \mathbf{n} = 0 \text{ on } \Gamma_{\text{up}} \text{ and } \Gamma_{\text{down}}, * \in \{\text{K, Ca, Mg, Al, C, Si, H}^+\} \quad (28-1)$$

$$(c_{\#} \vec{u} - \mathbf{D} \nabla c_{\#}) \bullet \mathbf{n} = 0 \text{ on } \Gamma_{\text{left}}, \# \in \{\text{K, Ca, Mg, Al, C, Si}\} \quad (28-2)$$

$$(c_{\text{H}^+} \vec{u} - \mathbf{D} \nabla c_{\text{H}^+}) \bullet \mathbf{n} = c_{\text{H}^+} \vec{u} \bullet \mathbf{n} \text{ on } \Gamma_{\text{left}} \quad (28-3)$$

$$\mathbf{D} \nabla c_{\text{H}^+} \bullet \mathbf{n} = 0 \text{ on } \Gamma_{\text{right}} \quad (28-4)$$

$$c_{\#} = 0 \text{ in } \Omega \quad (28-5)$$

$$c_{\text{H}^+} = 10^{-2} [\text{mol}/\text{m}^3] \text{ in } \Omega \quad (28-6)$$

The boundary conditions for Darcy's equation are set as follows:

$$\vec{u} \bullet \mathbf{n} = -1.5 \times 10^{-7} [\text{m}/\text{s}] \text{ on } \Gamma_{\text{left}} \quad (29-1)$$

$$\vec{u} \bullet \mathbf{n} = 0 \text{ on } \Gamma_{\text{up}} \text{ and } \Gamma_{\text{down}} \quad (29-2)$$

$$p = 0 \text{ on } \Gamma_{\text{right}} \quad (29-3)$$

5.2. Simulation results and discussion

In this simulation, calcium, magnesium, and carbon in the fluid come from the dissolution of the carbonate mineral dolomite, and the simulation results of the three elements in the fluid are similar, so only calcium is selected as the representative element of the carbonate dissolution for analysis. In addition, aluminum, potassium, and silicon in the fluid are derived from the dissolution of the silicate salt minerals quartz, K-feldspar, muscovite, and kaolinite. Similarly, only silicon is selected here as the representative element of the carbonate dissolution for analysis. Of course, the simulation results of other elements are shown in the supporting materials. Fig. 8 shows the simulation results of the calcium concentration, pH, and silicon concentration at different times. It can be seen from Fig. 8a and Fig. 8g that although the distribution forms of the calcium and silicon concentrations are the same, the concentration values of the two are more than 5 orders of magnitude different. This is because changes in the calcium concentration are closely related to the dissolution of dolomite (Eq. Eq. (19-2)), while changes in the silicon concentration are closely related to the dissolution of quartz, K-feldspar, muscovite, and kaolinite (Eq. (24)). It can be seen from Table 1 that under acidic conditions, the reaction rate constant of dolomite is more than 7 orders of magnitude higher than the reaction rate constant of the other four minerals, so the solute produced (or consumed) during the dissolution of dolomite is significantly greater than the solute produced by the dissolution of the other four minerals at the same time. In addition, as the simulation time progressed, the calcium concentration and the pH value changed significantly (Fig. 8a, 8b, 8c, 8d, 8e, and 8f), while the silicon concentration remained stable (Fig. 8g, 8h, and 8i). This is due to the fast reaction rate of dolomite, which makes it completely dissolve in the simulated period. With the complete dissolution of the dolomite upstream of the simulation domain, the calcium supply source is lost, and the calcium concentration is continuously reduced under the continuous washing of the upstream fluids. The change in the pH value is related to the dissolution of the dolomite, and after the dolomite is completely dissolved, the consumption of H^+ in the upstream mineral reaction process is also reduced, so the concentration of H^+ increases, that is, the pH value decreases. Because the dissolution rates of quartz, K-feldspar, muscovite, and kaolinite are very small, these four minerals still exist in large quantities throughout the simulation time, and there is no complete dissolution, so the silicon concentration remains stable. In addition, the distribution and trends of the carbon, magnesium, kalium, and aluminum concentrations can be found in the supplementary material.

Fig. 9a, 9b, and 9c show the relative changes in dolomite at different simulation times. In combination with Fig. 8, the complete dissolution of the dolomite is closely related to the changes in the calcium concentration and pH value, and the boundary line of the complete dissolution of dolomite is consistent with the contour line of the calcium concentration and the pH value. Fig. 9d, 9e, 9f, 9g, 9h, and 9i show that the amplitude of the variation in the dolomite is the same as that of the

Table 1
Parameters for calculating mineral kinetic rate constants.

Mineral	Parameters for kinetic rate law				
	$k_{25}^{\text{nu}} (\text{mol}/\text{m}^2/\text{s})$	$k_{25}^{\text{H}} (\text{mol}/\text{m}^2/\text{s})$	$n(\text{H} \pm)$	$k_{25}^{\text{OH}} (\text{mol}/\text{m}^2/\text{s})$	$n(\text{OH}-)$
Quartz	1.023E-14	—	—	—	—
Dolomite	2.951E-08	6.457E-04	0.5	—	—
K-feldspar	3.890E-13	8.710E-11	0.5	6.310E-22	-0.823
Kaolinite	6.918E-14	4.898E-12	0.777	8.913E-18	-0.472
Muscovite	3.020E-13	7.762E-12	0.5	—	—

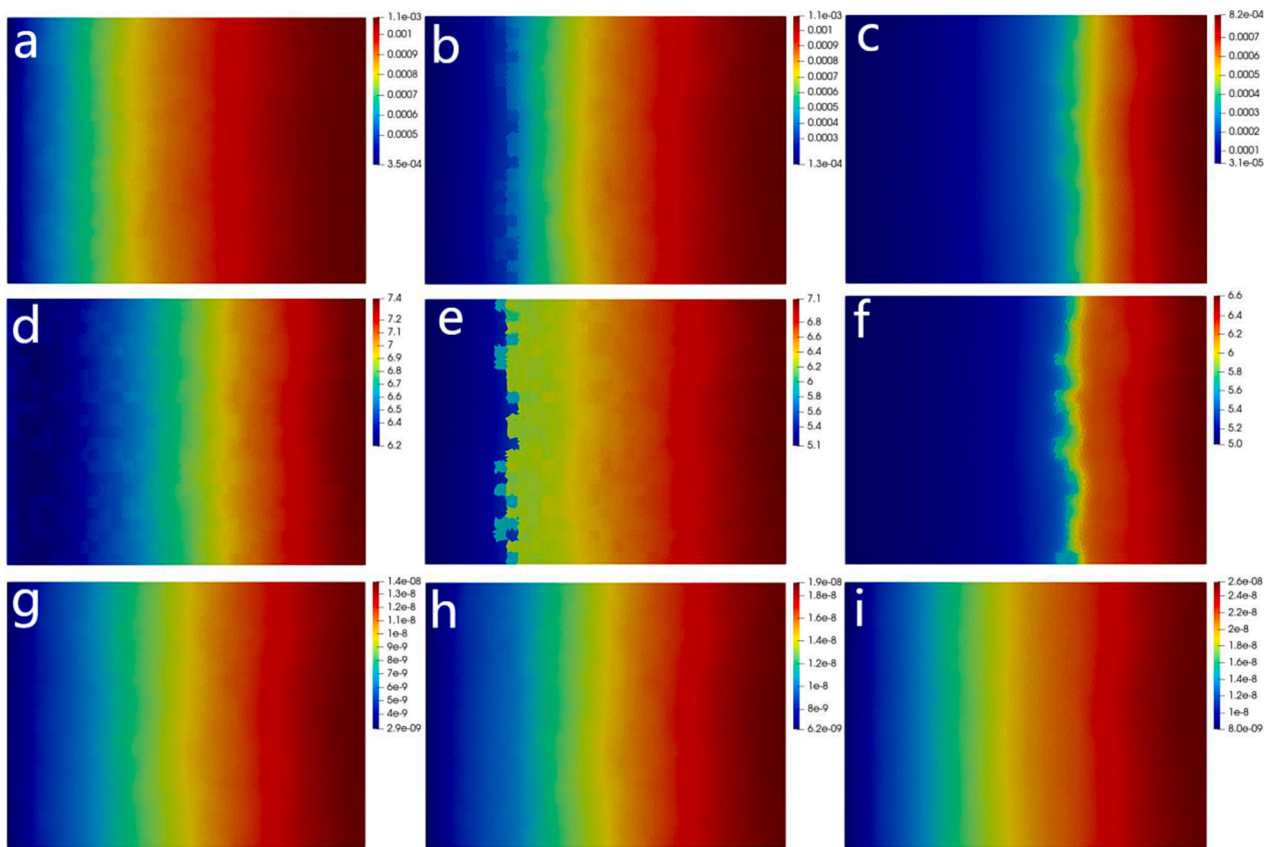


Fig. 8. Distribution of the calcium concentration [mol/L] (a, b, c), the pH value (d, e, f) and the silicon concentration [mol/L] (g, h, i) in the macroscopic simulation domain after simulation times of 4.32×10^5 s, 8.64×10^5 s and 1.296×10^6 s.

variation in the porosity. This feature will be better reflected if the variation in the dolomite and porosity is normalized. This indicates that the change in the porosity is completely dominated by the dissolution of dolomite, and the dissolution of silicate has little effect on the porosity, which is related to its very low dissolution rate (relative to carbonate minerals). The change in porosity usually causes a change in permeability and then also a change in the flow field. As shown in Fig. 10, with increasing porosity, the flow field becomes more uniform (Fig. 10a and 10b) and is closer to the inlet velocity (1.5×10^{-7} m/s) (Fig. 10c), which indicates that resistance to the flow velocity decreases with increasing porosity. Since the permeability distribution in the simulation domain is relatively uniform (Fig. 7d), the formation or strengthening of priority flow channels is not observed during the entire simulation process. In contrast, the existing fast flow zone becomes increasingly gentle.

To further understand the changes in the macroscopic effective parameters such as porosity, permeability tensor, diffusion tensor, and reaction specific surface area during the dissolution of dolomite, the geometric evolution data of some typical elements were selected for analysis. Fig. 11 shows the geometric evolution results of the 352nd REV in the simulation domain at different times, with a complete geometric evolution video (Movie S1) included in the [supplementary material](#). The four time nodes in Fig. 11 correspond one-to-one to the four time nodes marked (vertical dashed lines) in Fig. 12c. Fig. 12a shows that the porosity increases steadily during the dolomite dissolution process without an obvious inflection point, which is consistent with the dissolution process of dolomite (Fig. 11). Before the simulation time reaches the third time node, the permeability tensor and diffusion tensor steadily increase with increasing porosity, and the growth rate in the y-axis direction is greater than the growth rate in the x-axis direction, but the permeability tensor and diffusion tensor in the x-axis direction increase rapidly between the third and fourth time nodes, and the reverse

exceeds the values in the axis direction (Fig. 12a and 12b). Combined with Fig. 11c, it can be seen that this is due to the complete dissolution of the dolomite, which leads to the addition of an advantageous channel in the x-axis direction, making the flow and migration resistance in the x-axis direction much smaller.

It can be seen from Fig. 12c that before reaching the first time node, the growth rate of the reactive specific surface area of the four silicate minerals is the same because the reactive specific surface area (or fluid–solid interface) of the silicate minerals covered by the dissolved dolomite is the same during the period, and this can be seen even more in combination with Fig. 11a. After the first time node, the reactive surface areas of the kaolinite and the muscovite covered by the dolomite are completely exposed to the fluid (Fig. 11a), so the reactive specific surface areas of the kaolinite and the muscovite remain stable (Fig. 12c). Between the first and the second time nodes, the growth rate of the quartz's reactive specific surface area increased by a factor of three, while the growth rate of the K-feldspar remained unchanged (Fig. 12c). Combined with Fig. 11b, it can be seen that this is due to the evolution of the fluid–solid interface of dolomite on the left, which resulted in more quartz's reactive surface area being exposed to the fluid. The exposure rates of the quartz's and K-feldspar's reactive surface areas are unchanged due to the evolution of the fluid–solid interface of the right dolomite. After the first time node, the reactive surface area of K-feldspar covered by the dolomite is completely exposed to the fluid (Fig. 11b), so the reactive specific surface area of the K-feldspar remains stable for the remaining time. Between the second and the fourth time node, the growth rate of the quartz's reactive specific surface area is further increased, which can be seen in Fig. 11c, due to the evolution of the right dolomite fluid–solid interface, resulting in more quartz's reactive surface area being exposed to the fluid. Between the third time node and the fourth time node, the reactive specific surface area and the

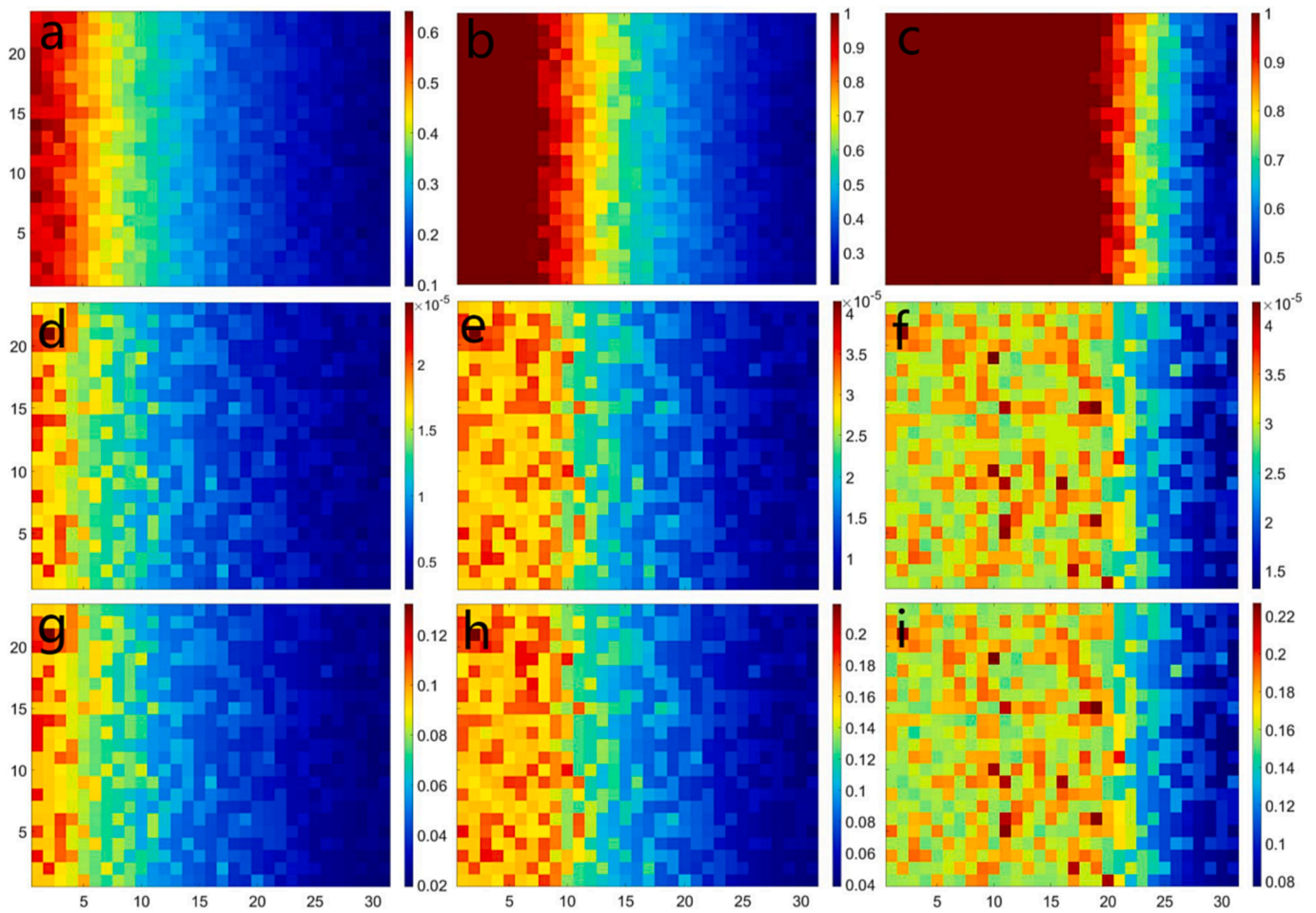


Fig. 9. Distribution of the relative change in the amount of dolomite (a, b, c), the absolute change in the amount of dolomite (d, e, f) [mol], and the absolute change in the amount of porosity (g, h, i) in REV's after simulation times 4.32×10^5 s, 8.64×10^5 s and 1.296×10^6 s.

total reactive specific surface area of the dolomite decreased, and the reactive specific surface area of the dolomite dropped to zero. Fig. 11c and Fig. 11d show that this is caused by the complete dissolution of the dolomite.

As shown in Fig. 12, macro parameters such as porosity remain constant after the fourth time node, indicating that the dissolution of the silicate does not lead to the obvious evolution of the typical unit geometry. Due to the very low dissolution rate of the silicates relative to the carbonate (the difference is more than five orders of magnitude), our model is unable to capture the fluid–solid interface evolution of both on the same spatial and temporal scales. However, surprisingly, the content of silicate minerals still changed during the simulation process. It is not difficult to see that the volume of silicate minerals did change by comparing Fig. 11 and Fig. 7b. Therefore, we consider this change is not real, but due to the error that is caused by the change in the dolomite volume, and this error is related to the numerical algorithm we used the VIIM, which retains only first-order accuracy. In fact, a second-order discretized spatial gradients can be used to accurately characterize geometric evolution, as demonstrated in Gärttner et al. (2020). Doing so not only allows a more accurate prediction of the volume of mineral solids, but also an accurate representation of its shape. The reason we did not use this approach to improve the VIIM is that higher-order correction would have greatly increased the computational effort. Therefore, we consider that if the first-order precision can meet the accuracy requirements, there is no need to spend more computational effort. This error must be quantified to assess the reliability (or accuracy) of the simulation results. Here we propose two methods to quantify the errors arising from the geometric evolution process. The first method is

to quantify the difference between the specific surface area of the silicate reaction and the real value at the end of the simulation. This method is only applicable to the REV's where the dolomite is completely dissolved, it is not universal and it cannot evaluate the error caused by the dolomite. The second method is to calculate the error in the simulation process via mass conservation, which is suitable for all cases. In this paper, we use the second method. First, we know that the silicate mineral content hardly changes during the simulation time, and the change in the pore volume is completely caused by the dissolution of the dolomite, so the change in the pore volume should be equal to the change in the dolomite volume. The relative error caused by the change in the dolomite volume is defined by the following method:

$$\text{Error}_{\text{Dolomite}} = \frac{|\Delta\phi - \Delta V_{\text{Dolomite}}|}{\Delta\phi} \cdot P_{\text{Dolomite}} \times 100\% \quad (30)$$

where $\Delta\phi$ is the change in the porosity before and after the simulation, $\Delta V_{\text{Dolomite}}$ is the change in the dolomite volume fraction, and P_{Dolomite} is the volume fraction of the solid skeleton occupied by the dolomite. The relative error generated by the four silicate minerals is the relative change before and after their simulation:

$$\text{Error}^* = \frac{|\Delta M^*|}{M_0} \times 100\%, * \in \{\text{K.feldspar, Kaolinite, Muscovite, Quartz}\} \quad (31)$$

where ΔM is the change in the mineral amount before and after simulation, and M_0 is the mineral content at the initial moment.

According to Eqs. (30) and (31), we calculate the relative errors

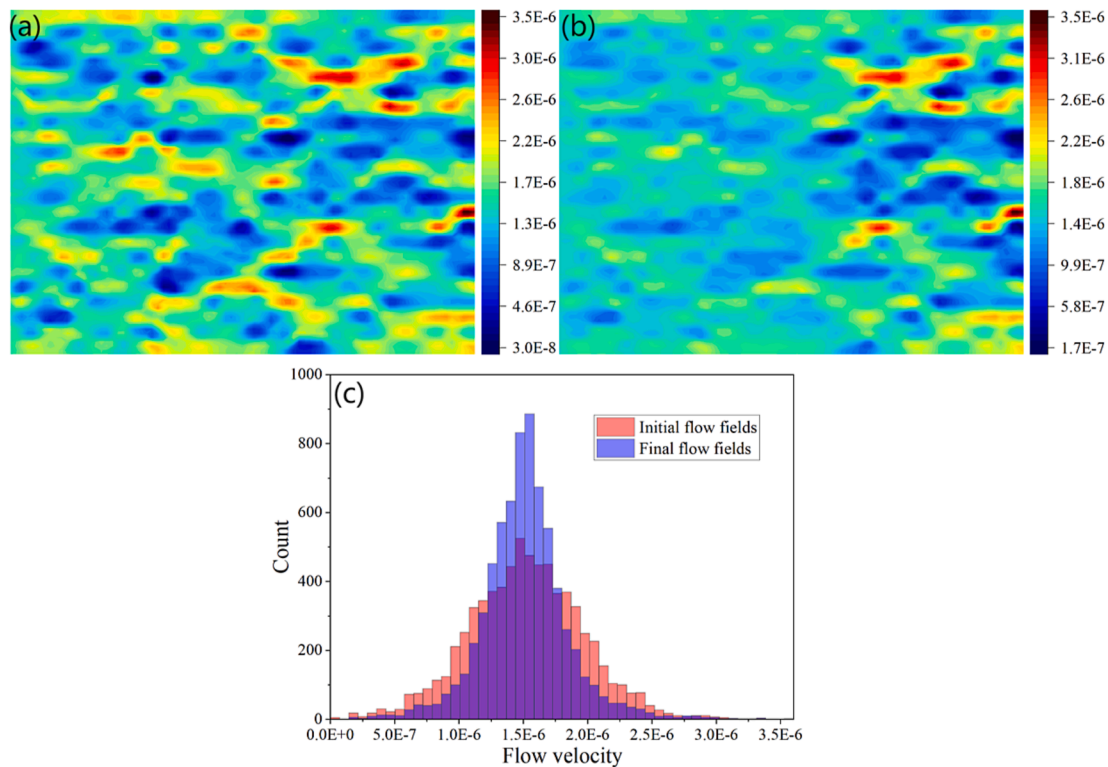


Fig. 10. Flow velocity distribution at the initial moment (a) and at the end of the simulation (b) [dm/s], (c) flow velocity histogram.

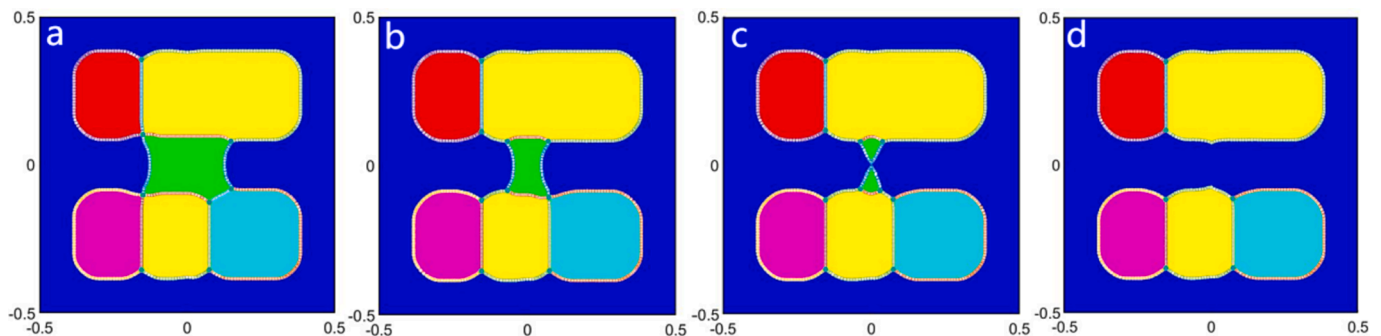


Fig. 11. Evolution results of the fluid–solid interface of the 352nd REV (row 12 column 11) after simulation times of 7.61×10^5 s (a), 9.11×10^5 s (b), 9.63×10^5 s (c) and 1.003×10^6 s (d).

generated by all the typical elements, and the results of the calculation for various minerals can be found in the [supplementary material](#). The mean relative errors of dolomite, K-feldspar, kaolinite, muscovite, and quartz are 1.80 %, 3.03 %, 4.96 %, 4.96 %, and 0.99 %, respectively. The relative errors of each mineral are negatively correlated with its volume fraction. By calculating the weighted average, we obtain a total mean relative error of 2.39 %, which can approximately be considered mass conservation. Therefore, we conclude that VIIM is suitable for dealing with the complex geometric evolution of typical units. In addition, the model runs on a computer with an Intel Core i9-12900 K CPU, utilizing 16 physical cores for an overall computation time of approximately 33,000 s. The models all occupy less than 32 GB of memory at runtime.

6. Conclusion

Based on the RTSPHEM model framework and coupled with PHREEQC software, we developed a micro–macro model for multi-component reactive transport in multimineral systems (RTSPHEM-P). The model can effectively handle challenging geometric settings, which

often result in the emergence of many triple points and complex topological changes. Through the simulation of the same problem, the RTSPHEM-P model shows more reasonable simulation results than the original model framework, which is due to its advanced chemistry solver. Through the direct modeling and simulation of real SEM images, some unique model capabilities are demonstrated, such as capturing the abrupt changes in the permeability and diffusion tensors that are caused by geometric topological changes and characterizing the evolution of their reaction specific surface area under multimineral cementation. By checking the simulation results and analyzing the error, our method has proven to be stable and has a good approximate mass conservation effect.

Although all of the minerals in the simulation exhibit dissolution properties, the model is capable of handling mineral precipitation, provided that no more than two minerals are precipitated simultaneously. This is because when multiple minerals are simultaneously precipitated, the model will face the uncertainty of the interface precipitation site, which the model cannot deal with effectively. Our multimineral model is formulated for systems that contain at most one

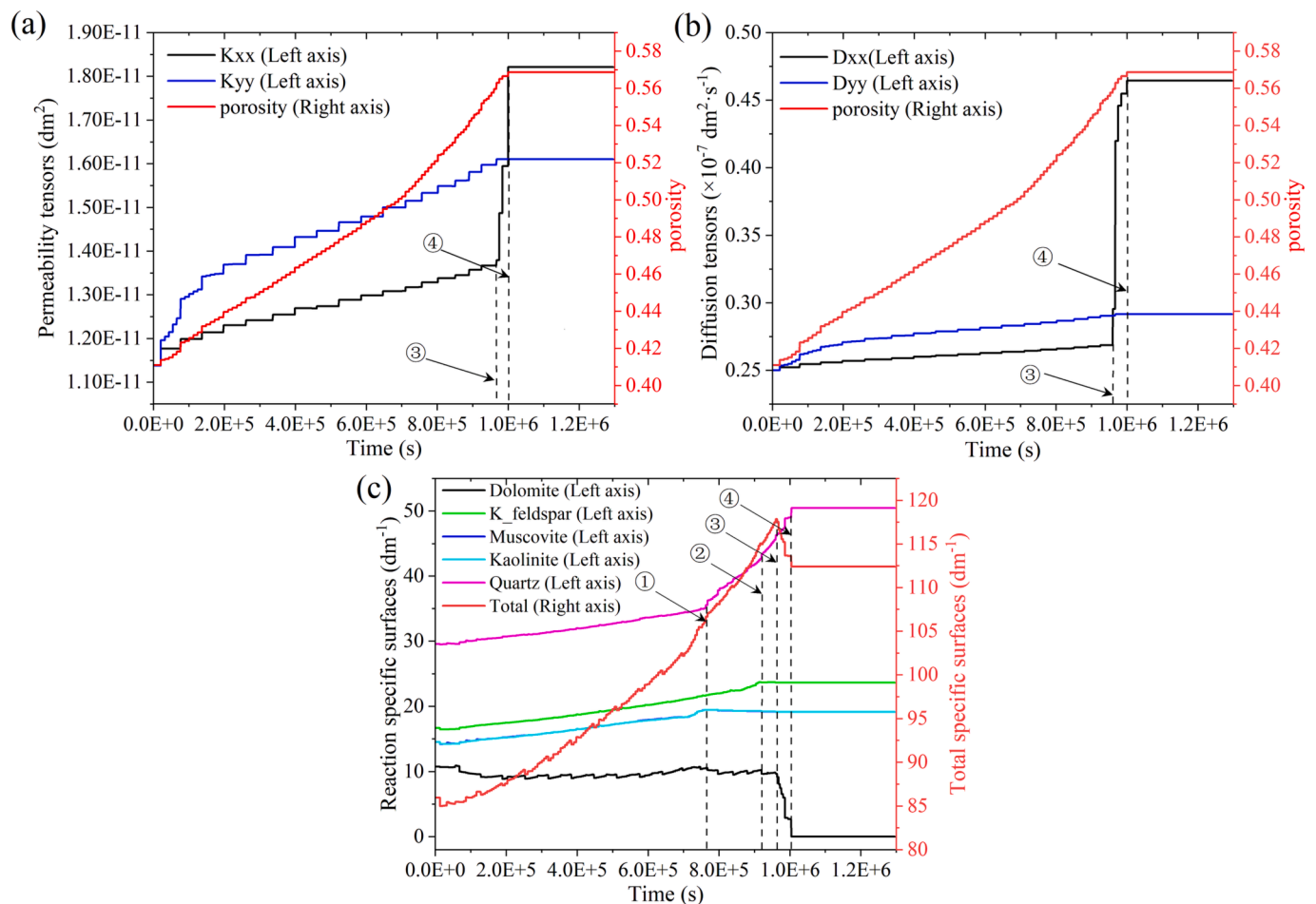


Fig. 12. Diachronic changes in porosity, permeability tensor, diffusion tensor, and reaction specific surface area during the geometric evolution of the 352nd REV.

precipitated mineral, so further research is necessary to weaken these restrictions in the geometric evolution. At present, an effective strategy is to add a probability density function to characterize multiminerall precipitation at the fluid–solid interface. In addition, the VIIM for geometric evolution exhibits only first-order accuracy and is unable to accurately capture relatively fine interface movements, and further research is needed to find strategies to remedy this drawback.

Since the basic methods applied in this paper are not limited to two spatial dimensions or to smaller spatial scales, scale separation (i.e., micro- and macro-scales can be described independently of each other), our model is very suitable for generalizing larger-scale site problems, such as CO₂ geological sequestration and the formation of underground cave systems.

CRediT authorship contribution statement

Heng Li: Writing – original draft, Methodology, Conceptualization. **Fugang Wang:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization. **Qingcheng He:** Writing – review & editing, Project administration, Funding acquisition. **Yaohui Wang:** Visualization, Validation, Software. **Yilong Yuan:** Validation, Supervision, Project administration, Methodology. **Guanhong Feng:** Resources, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (Grant Nos. U2244215, 42072280) and Graduate Innovation Fund of Jilin University. The study was also supported by Key Laboratory of Groundwater Resources and Environment (Jilin University), Ministry of Education, and by Jilin Provincial Key Laboratory of Water Resources and Environment, Jilin University.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2024.130764>.

References

- Allaire, G., 1992. Homogenization and two-scale convergence. *SIAM J. Math. Anal.* 23 (6), 1482–1518. <https://doi.org/10.1137/0523084>.
- Amanbek, Y., Singh, G., Wheeler, M.F., van Duijn, H., 2019. Adaptive numerical homogenization for upscaling single phase flow and transport. *J. Comput. Phys.* 387, 117–133. <https://doi.org/10.1016/j.jcp.2019.02.014>.
- Auriault, J.-L., Boutin, C., Geindreau, C., 2009. Homogenization of Coupled Phenomena in Heterogeneous Media. ISTE Ltd and John Wiley & Sons, Inc, London and Hoboken, 465 pp doi: 10.1002/9780470612033.
- Ball, J.W., Nordstrom, D., 1991. WATEQ4F - User's manual with revised thermodynamic data base and test cases for calculating speciation of major, trace and redox elements in natural waters. U.S. Geological Survey Open-File Report 91–183, 1–189.

- Bastidas, M., Bringedal, C., Pop, I.S., 2021. Numerical Simulation of a Phase-Field Model for Reactive Transport in Porous Media. In: Vermolen, F.J., Vuik, C. (Eds.), *Numerical Mathematics and Advanced Applications ENUMATH 2019*. Springer International Publishing, Cham, pp. 93–102. https://doi.org/10.1007/978-3-030-55874-1_8.
- Bensoussan, A., Lions, J.-L., Papanicolaou, G., 1978. *Asymptotic Analysis for Periodic Structures*. Studies in Mathematics and Its Applications, 5. Elsevier, New York, p. 700.
- Bourbatache, M.K., Hellou, M., Lominé, F., 2019. Two-scale analysis of the permeability of 3D periodic granular and fibrous media. *Acta Mech.* 230 (10), 3703–3721. <https://doi.org/10.1007/s00707-019-02470-3>.
- Bourbatache, K., Millet, O., Ait-Mokhtar, A., 2012. Ionic transfer in charged porous media. Periodic homogenization and parametric study on 2D microstructures. *Int. J. Heat Mass Transf.* 55 (21), 5979–5991. <https://doi.org/10.1016/j.ijheatmasstransfer.2012.06.008>.
- Bourbatache, K., Millet, O., Ait-Mokhtar, A., Amiri, O., 2013. Chloride transfer in cement-based materials. Part 1. Theoretical basis and modelling. *Int. J. Numer. Anal. Meth. Geomech.* 37 (11), 1614–1627. <https://doi.org/10.1002/nag.2102>.
- Bourbatache, K., Millet, O., Ait-Mokhtar, A., 2016. Multi-scale periodic homogenization of ionic transfer in cementitious materials. *Heat Mass Transf.* 52 (8), 1489–1499. <https://doi.org/10.1007/s00231-015-1667-3>.
- Bourbatache, M.K., Millet, O., Moyné, C., 2020. Upscaling diffusion–reaction in porous media. *Acta Mech.* 231 (5), 2011–2031. <https://doi.org/10.1007/s00707-020-02631-9>.
- Bringedal, C., Berre, I., Pop, I.S., Radu, F.A., 2016a. Upscaling of non-isothermal reactive porous media flow with changing porosity. *Transp. Porous Media* 114 (2), 371–393. <https://doi.org/10.1007/s11242-015-0530-9>.
- Bringedal, C., Berre, I., Pop, I.S., Radu, F.A., 2016b. Upscaling of nonisothermal reactive porous media flow under dominant péclet number: the effect of changing porosity. *Multiscale Model. Simul.* 14 (1), 502–533. <https://doi.org/10.1137/15M1022781>.
- Bringedal, C., Kumar, K., 2017. Effective behavior near clogging in upscaled equations for non-isothermal reactive porous media flow. *Transp. Porous Media* 120 (3), 553–577. <https://doi.org/10.1007/s11242-017-0940-y>.
- Bringedal, C., von Wolff, L., Pop, I.S., 2020. Phase field modeling of precipitation and dissolution processes in porous media: upscaling and numerical experiments. *Multiscale Modeling & Simulation* 18 (2), 1076–1112. <https://doi.org/10.1137/19M1239003>.
- Chew, A.W.Z., Law, A.-W.-K., 2020. Homogenization theory with multiscale perturbation analysis for supervised learning of complex adsorption-desorption process in porous-media systems. *Journal of Computational Science* 40, 101071. <https://doi.org/10.1016/j.jocs.2019.101071>.
- Christie, M.A., Blunt, M.J., 2001. Tenth SPE comparative solution project: a comparison of upscaling techniques, SPE Reservoir Simulation Symposium, pp. SPE-66599-MS. doi: 10.2118/66599-MS.
- Ebigbo, A., Helmig, R., Cunningham, A.B., Class, H., Gerlach, R., 2010. Modelling biofilm growth in the presence of carbon dioxide and water flow in the subsurface. *Adv. Water Resour.* 33 (7), 762–781. <https://doi.org/10.1016/j.advwatres.2010.04.004>.
- Frolković, P., Gajdošová, N., Gärttner, S., Ray, N., 2020. Voronoi implicit interface method for geometry evolution of two minerals with applications in reactive porous media. In: *Proceedings of the Conference Algorithm*, pp. 121–130.
- Gärttner, S., Frolković, P., Knabner, P., Ray, N., 2020. Efficiency and accuracy of micro-macro models for mineral dissolution. *Water Resources Research*, 56(8): e2020WR027585 doi: 10.1029/2020WR027585.
- Gärttner, S., Frank, F., 2021. *RTSPHEM – Reactive Transport Solver in Porous Homogenized Evolving Media*. Friedrich-Alexander University Erlangen-Nürnberg, Erlangen <https://github.com/cupperfreeze/RTSPHEM/>.
- Gärttner, S., Frolković, P., Knabner, P., Ray, N., 2022. Efficiency of micro-macro models for reactive two-mineral systems. *Multiscale Model. Simul.* 20 (1), 433–461. <https://doi.org/10.1137/20M1380648>.
- Hornung, U., 1997. *Homogenization and Porous Media*. Interdisciplinary Applied Mathematics. Springer New York, 1-295 pp doi: 10.1007/978-1-4612-1920-0.
- Kumar, K., van Noorden, T.L., Pop, I.S., 2011. Effective dispersion equations for reactive flows involving free boundaries at the microscale. *Multiscale Model. Simul.* 9 (1), 29–58. <https://doi.org/10.1137/100804553>.
- Lasaga, A.C., 1984. Chemical kinetics of water-rock interactions. *J. Geophys. Res. Solid Earth* 89 (B6), 4009–4025. <https://doi.org/10.1029/JB089iB06p04009>.
- Lasaga, A.C., Soler, J.M., Ganor, J., Burch, T.E., Nagy, K.L., 1994. Chemical weathering rate laws and global geochemical cycles. *Geochim. Cosmochim. Acta* 58 (10), 2361–2386. [https://doi.org/10.1016/0016-7037\(94\)90016-7](https://doi.org/10.1016/0016-7037(94)90016-7).
- Li, H., 2023. RTSPHEM-P: July 19, 2023 Release (Version 2) [Software]. Zenodo. <https://doi.org/10.5281/zenodo.8163785>.
- Li, X., Huang, H., Meakin, P., 2008. Level set simulation of coupled advection-diffusion and pore structure evolution due to mineral precipitation in porous media. *Water Resour. Res.* 44 (12) <https://doi.org/10.1029/2007WR006742>.
- Lopes, T.V., et al., 2020. A new computational model for flow in karst-carbonates containing solution-collapse breccias. *Comput. Geosci.* 24 (1), 61–87. <https://doi.org/10.1007/s10596-019-09894-9>.
- Ma, J., Ahkami, M., Saar, M.O., Kong, X.-Z., 2021. Quantification of mineral accessible surface area and flow-dependent fluid-mineral reactivity at the pore scale. *Chem. Geol.* 563, 120042 <https://doi.org/10.1016/j.chemgeo.2020.120042>.
- MATLAB, 2021. Version 9.11.0.1769968. The MathWorks Inc, Natick, Massachusetts.
- Metzger, S., Knabner, P., 2021. Homogenization of two-phase flow in porous media from pore to darcy scale: a phase-field approach. *Multiscale Model. Simul.* 19 (1), 320–343. <https://doi.org/10.1137/19M1287705>.
- Molins, S., Trebotich, D., Miller, G.H., Steefel, C.I., 2017. Mineralogical and transport controls on the evolution of porous media texture using direct numerical simulation. *Water Resour. Res.* 53 (5), 3645–3661. <https://doi.org/10.1002/2016WR020323>.
- Molins, S., Trebotich, D., Arora, B., Steefel, C.I., Deng, H., 2019. Multi-scale model of reactive transport in fractured media: diffusion limitations on rates. *Transp. Porous Media* 128 (2), 701–721. <https://doi.org/10.1007/s11242-019-01266-2>.
- Muntean, A., Nikolopoulos, C., 2020. Colloidal transport in locally periodic evolving porous media—an upscaling exercise. *SIAM J. Appl. Math.* 80 (1), 448–475. <https://doi.org/10.1137/17M1161531>.
- Olsson, E., Kreiss, G., 2005. A conservative level set method for two phase flow. *J. Comput. Phys.* 210 (1), 225–246. <https://doi.org/10.1016/j.jcp.2005.04.007>.
- Olsson, E., Kreiss, G., Zahedi, S., 2007. A conservative level set method for two phase flow II. *J. Comput. Phys.* 225 (1), 785–807. <https://doi.org/10.1016/j.jcp.2006.12.027>.
- Osher, S., Fedkiw, R., 2003. *Level Set Methods and Dynamic Implicit Surfaces*. Applied Mathematical Sciences. Springer, New York, p. 153.
- Palandri, J., Kharaka, Y., 2004. A compilation of rate parameters of water-mineral interaction kinetics for application to geochemical modeling. 2004-1068, U.S. Geological Survey, U.S. Geological Survey doi: 10.3133/ofr20041068.
- Parkhurst, D.L., Appelo, C.A.J., 2013. Description of input and examples for PHREEQC version 3—A computer program for speciation, batch-reaction, one-dimensional transport, and inverse geochemical calculations. U.S. Geological Survey.
- Ping, S., et al., 2023. Multi-scale deterioration mechanism of shear strength of gypsum-bearing mudstone induced by water-rock reactions. *Eng. Geol.* 323, 107224 <https://doi.org/10.1016/j.enggeo.2023.107224>.
- Ray, N., van Noorden, T., Frank, F., Knabner, P., 2012. Multiscale modeling of colloid and fluid dynamics in porous media including an evolving microstructure. *Transp. Porous Media* 95 (3), 669–696. <https://doi.org/10.1007/s11242-012-0068-z>.
- Ray, N., van Noorden, T., Radu, F.A., Friess, W., Knabner, P., 2013. Drug release from collagen matrices including an evolving microstructure. *ZAMM - Journal of Applied Mathematics and Mechanics / Zeitschrift Für Angewandte Mathematik Und Mechanik* 93 (10–11), 811–822. <https://doi.org/10.1002/zamm.201200196>.
- Ray, N., Oberlander, J., Frolkovic, P., 2019. Numerical investigation of a fully coupled micro-macro model for mineral dissolution and precipitation. *Comput. Geosci.* 23 (5), 1173–1192. <https://doi.org/10.1007/s10596-019-09876-x>.
- Saye, R.L., Sethian, J.A., 2012. Analysis and applications of the Voronoi Implicit Interface Method. *Journal of Computational Physics* 231 (18), 6051–6085. <https://doi.org/10.1016/j.jcp.2012.04.004>.
- Scheibe, T.D., et al., 2015. An analysis platform for multiscale hydrogeologic modeling with emphasis on hybrid multiscale methods. *Groundwater* 53 (1), 38–56. <https://doi.org/10.1111/gwat.12179>.
- Schulz, R., Knabner, P., 2017a. Derivation and analysis of an effective model for biofilm growth in evolving porous media. *Mathematical Methods in the Applied Sciences* 40 (8), 2930–2948. <https://doi.org/10.1002/mma.4211>.
- Schulz, R., Knabner, P., 2017b. An effective model for biofilm growth made by chemotactical bacteria in evolving porous media. *SIAM J. Appl. Math.* 77 (5), 1653–1677. <https://doi.org/10.1137/16M108817X>.
- Schulz, R., Ray, N., Frank, F., Mahato, H.S., Knabner, P., 2017. Strong solvability up to clogging of an effective diffusion–precipitation model in an evolving porous medium. *Eur. J. Appl. Math.* 28 (2), 179–207. <https://doi.org/10.1017/S0956792516000164>.
- Steefel, C.I., et al., 2014. Reactive transport codes for subsurface environmental simulation. *Comput. Geosci.* 19 (3), 445–478. <https://doi.org/10.1007/s10596-014-9443-x>.
- Steefel, C.I., 2019. Reactive transport at the crossroads. *Rev. Mineral. Geochem.* 85 (1), 1–26. <https://doi.org/10.2138/rmg.2019.85.1>.
- Steefel, C.I., MacQuarrie, K.T.B., 1996. Approaches to modeling of reactive transport in porous media. *Rev. Mineral. Geochem.* 34 (1), 85–129. <https://doi.org/10.1515/9781501509797-005>.
- Steefel, C.I., Molins, S., Trebotich, D., 2013. Pore scale processes associated with subsurface CO₂ injection and sequestration. *Rev. Mineral. Geochem.* 77 (1), 259–303. <https://doi.org/10.2138/rmg.2013.77.8>.
- Steefel, C.I., Beckingham, L.E., Landrot, G., 2015. Micro-continuum approaches for modeling pore-scale geochemical processes. *Rev. Mineral. Geochem.* 80 (1), 217–246. <https://doi.org/10.2138/rmg.2015.80.07>.
- Steefel, C.I., Lasaga, A.C., 1994. A coupled model for transport of multiple chemical species and kinetic precipitation/dissolution reactions with application to reactive flow in single phase hydrothermal systems. *Am. J. Sci.* 294 (5), 529. <http://www.ajsonline.org/content/294/5/529.abstract>.
- Tartakovsky, A.M., Tartakovsky, D.M., Scheibe, T.D., Meakin, P., 2008. Hybrid simulations of reaction-diffusion systems in porous media. *SIAM J. Sci. Comput.* 30 (6), 2799–2816. <https://doi.org/10.1137/070691097>.
- Tomin, P., Lunati, I., 2013. Hybrid multiscale finite volume method for two-phase flow in porous media. *J. Comput. Phys.* 250, 293–307. <https://doi.org/10.1016/j.jcp.2013.05.019>.
- Tournassat, C., Steefel, C.I., 2019. Reactive transport modeling of coupled processes in nanoporous media. *Rev. Mineral. Geochem.* 85 (1), 75–109. <https://doi.org/10.2138/rmg.2019.85.4>.
- van Noorden, T.L., 2009a. Crystal precipitation and dissolution in a porous medium: effective equations and numerical experiments. *Multiscale Model. Simul.* 7 (3), 1220–1236. <https://doi.org/10.1137/080722096>.
- van Noorden, T.L., Pop, I.S., Ebigbo, A., Helmig, R., 2010. An upscaled model for biofilm growth in a thin strip. *Water Resour. Res.* 46 (6) <https://doi.org/10.1029/2009WR008217>.

- Van Noorden, T.L., 2009b. Crystal precipitation and dissolution in a thin strip. *European Journal of Applied Mathematics*, 20(1): 69–91 <https://doi.org/10.1017/S0956792508007651>.
- Weinan, E., 2011. *Principles of Multiscale Modeling*. Cambridge University Press, New York.
- Whitaker, S., 1999. The method of volume averaging. theory and applications of transport in porous media, 13. Springer Netherlands, Dordrecht, XVI, 210 pp doi: 10.1007/978-94-017-3389-2.
- Wu, H., Zhao, J., Guo, N., 2019. Multiscale modeling of compaction bands in saturated high-porosity sandstones. *Eng. Geol.* 261, 105282 <https://doi.org/10.1016/j.enggeo.2019.105282>.
- Xu, T., Apps, J.A., Pruess, K., 2004. Numerical simulation of CO₂ disposal by mineral trapping in deep aquifers. *Appl. Geochem.* 19 (6), 917–936. <https://doi.org/10.1016/j.apgeochem.2003.11.003>.
- Xu, Z., Huang, H., Li, X., Meakin, P., 2012. Phase field and level set methods for modeling solute precipitation and/or dissolution. *Comput. Phys. Commun.* 183 (1), 15–19. <https://doi.org/10.1016/j.cpc.2011.08.005>.
- Yao, W., et al., 2023. Multi-scale deterioration of physical and mechanical properties of argillaceous siltstone under cyclic wetting-drying of Yangtze River water. *Eng. Geol.* 312, 106925 <https://doi.org/10.1016/j.enggeo.2022.106925>.
- Yeh, G.T., Tripathi, V.S., 1989. A critical evaluation of recent developments in hydrogeochemical transport models of reactive multichemical components. *Water Resour. Res.* 25 (1), 93–108. <https://doi.org/10.1029/WR025i001p00093>.
- Zhang, X., et al., 2021. Application of upscaling methods for fluid flow and mass transport in multi-scale heterogeneous media: a critical review. *Appl. Energy* 303, 117603. <https://doi.org/10.1016/j.apenergy.2021.117603>.
- Zhang, D., Zhang, X., Du, W., 2022. DEM-FEM based numerical analysis on mechanical responses of sandy soil and pipeline to seepage erosion. *Eng. Geol.* 310, 106868 <https://doi.org/10.1016/j.enggeo.2022.106868>.